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Enhancement of Machine Learning Models for Urban Heat Island Assessment

Master's Thesis

Author: Alexey Chechin

UT supervisor: Dr. Ir. Monica Pena Acosta

Promoter: Dr. Ir. Farid Vahdatikhaki

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1 Introduction

This Master's thesis explores the enhancement of machine learning models for Urban Heat Island (UHI) assessment, with a focus on capturing spatial dependencies that are often overlooked in traditional approaches. The Urban Heat Island effect, characterised by elevated temperatures in urban areas compared to their rural surroundings, poses increasing risks to public health, infrastructure, and urban resilience—particularly in the context of climate change and growing urban populations.

The thesis begins by introducing the concept of UHI and outlining the motivation behind improving data-driven assessment models. It then identifies a gap in the existing literature: the lack of spatially-aware modelling techniques that consider the influence of surrounding urban forms on local heat intensity. This forms the basis for the research problem, which is addressed through a combination of methodological innovation and empirical analysis. Following the problem definition, a detailed research design is presented, including clearly formulated objectives, research questions, and a justification for the selected methodologies. The thesis proceeds with a comprehensive literature review, covering relevant machine learning techniques, spatial regression approaches, and recent advances in spatially-informed prediction models.

Subsequently, the methodology chapter outlines the process of data preparation, spatial feature engineering, model development, and evaluation. Several machine learning models—including Random Forests and neural networks—are tested and compared, both with and without spatial enhancements. The results are analysed in detail in the discussion chapter, where findings are interpreted in the context of the research questions, and the strengths and limitations of the approach are critically evaluated. Finally, the conclusion chapter summarises the main contributions of the study and suggests potential directions for further development in spatially-informed environmental modelling.

1.1 Problem Analysis

In the current section, firstly a broader context of the problem is outlined by exploring the overarching challenge, previous attempts to approach it and the unaddressed research gap. Secondly, the problem is analysed in terms of its dimensions, followed by formulating a problem statement.

1.1.1 Problem Context

Natural weather events of excessive heat or abrupt temperature change result annually in over 702 deaths in the United States of America, with a total of 10,527 lethal cases registered between 2004 and 2018. [1] Indeed, the daytime temperature of inner-city urban areas rises up to 7 °C higher than that of outer-city regions, which research predicts to increase further in magnitude in the future due to increasing population density and climate change. [2]

One of the major causes of these challenges is a phenomenon called Urban Heat Island (UHI). UHI is defined as the difference between the temperature of the rural environment and the inner cities. Its main driver is the heat retained in impermeable man-made surfaces such as concrete or asphalt. [3] Therefore, it is essential to develop robust approaches to support urban design solutions that mitigate and prevent the UHI in future urban development projects.

Extensive research has been conducted on the Urban Heat Island (UHI) phenomenon, guided by two major modelling philosophies: physics-based simulations and data-driven models [3]. Although robust simulation-based tools exist, they require detailed domain knowledge and comprehensive input data about the studied area, which limits their practical applicability for planners and practitioners [4].

An alternative approach is data-driven modelling, in which algorithms identify patterns in large datasets to support decision-making in complex environments [5]. Compared to physics-based methods, data-driven models are typically faster to develop, easier to scale, and more adaptable to diverse urban contexts with limited input information. They can leverage publicly available data sources and offer actionable insights without requiring extensive simulation calibration. Since the goal of this work is to provide a modelling approach that is both interpretable and practical for informing UHI mitigation strategies, the focus in the remainder of this thesis is on data-driven modelling techniques.

Predominantly, the existing literature on data-driven solutions focuses on modelling the cities as a whole, which leads to excessive generalisation and locally irrelevant or counter-effective solutions. [6] However, to mitigate the consequences of the UHI effectively, decision-makers need to be facilitated with not only science-based but also specific context strategies. Some authors made steps towards a more urban context-specific approach to UHI. For example, Stewart and Oke [7] introduced a classification scheme to split the urban landscape into zones with comparable climatic behaviour.

Continuing this research line, Pena Acosta [3] focused on developing a user-friendly data-driven UHI assessment tool that focuses on the urban context-specific features on the street level. Moreover, Pena Acosta [3] explored two distinct effects that have an impact on human health and urban temperatures: Surface Urban Heat Island (SUHI) and Canopy Urban Heat Island (CUHI). The author also proposed a substantial enhancement in data collection by showing that the interplay of these two effects can either amplify or mitigate heat stress, which makes it important to analyse both types separately. [3]

Despite the significant progress in data-driven assessment of the UHI phenomenon described above, there are several major challenges that researchers face. Namely, in the study of Pena Acosta [3], it is admitted that while features such as building height and street width are independent, the cumulative effect of heat captured and emitted by one area influences nearby locations. This spatial interaction is what the current models fail to capture, which is a significant disadvantage since in reality the microclimate is highly influenced by interactions at the boundaries of different areas. In other words, while in reality, the temperature propagation is a highly dynamic spatio-temporal process, the current data-driven approaches limit the model to a static representation by not including spatial dependencies as predictors. Therefore, upon the request of the problem owner, the current paper focuses further on addressing the lack of spatial interdependency between features based on the UHI assessment models of Pena Acosta [3].

1.1.2 Problem Formulation

As mentioned in the previous section, the main challenge addressed in the designed research is the lack of ability of the current data-driven models to incorporate the spatial interdependency of features. Indeed, the research of Pena Acosta [3] assigns a range of urban morphological and socio-economic parameters to each street segment based on the buffers created around each piece of the street, as can be seen in Figure 1. Since the microclimate has a fluid and dynamic nature, it can be expected that the spatial interplay of the parameters at the boundaries of the buffers has a significant influence on how the UHI pattern is formed.

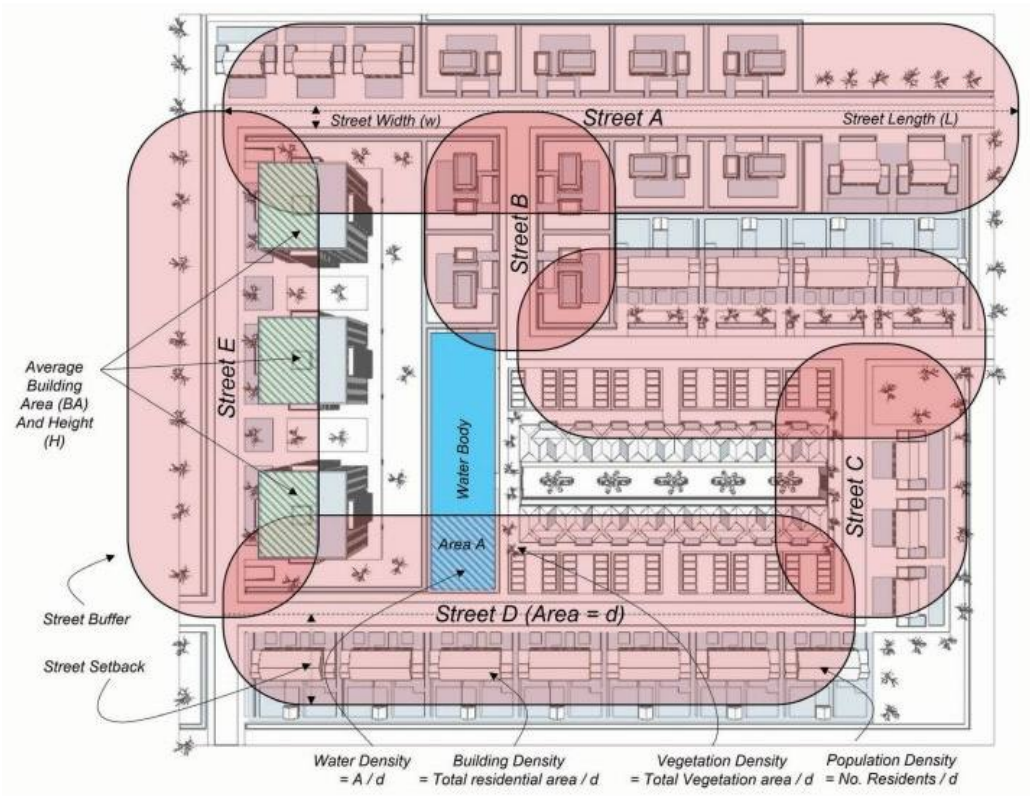


Figure 1: Buffer zones [3]

Moreover, the performance of the existing UHI assessment models suggests that substantial improvements are possible. In their study, Pena Acosta [3] used traditional machine learning models such as Random Forest (RF) and Artificial Neural Network (ANN). However, the spatial constituent of data was not present in the training set, which gives this modelling approach a static nature. It was concluded that the performance of such models ranges between 0.75 and 0.82 R^2 values for the Artificial Neural Networks and between 0.80 and 0.83 for the

Random Forest algorithm, depending on the chosen indicator. [3] It can therefore be supposed that the reason for 17% to 25% of variability in data not being captured by the developed ANN and RF models respectively can potentially originate from the absence of spatial component in the data. Therefore, Pena Acosta [3] suggests that it is necessary to develop a model that would incorporate spatial dependencies between the data entries.

Simply put, the Urban Heating Island assessment models are currently based on 3 pillars as input: socio-economic features, urban morphological features and temperature measurements of both outer (reference) and inner city. However, according to the hypothesis made in the current thesis, what is missing in the data to represent the nature of heat propagation is the spatial distribution of the street segments to which all the other features are attributed. The conceptual overview of the dimensions of the problem and the focus of the present research explained above can be seen in Figure 2.

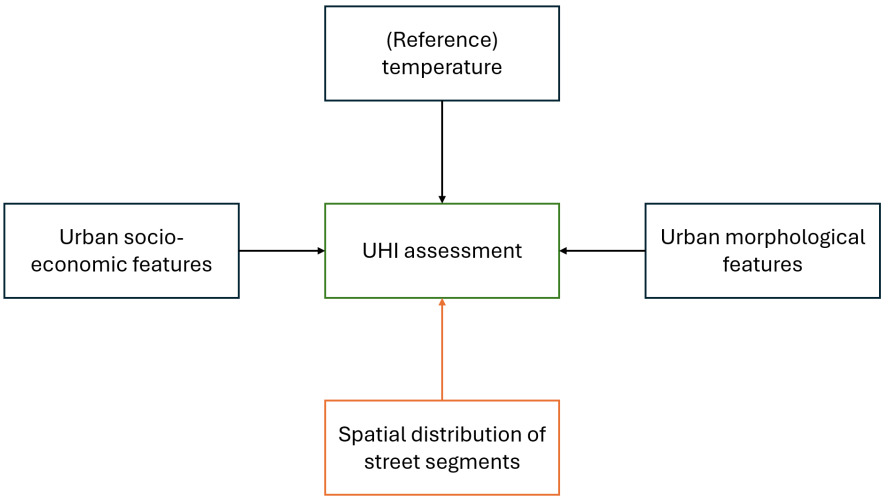


Figure 2: Problem Analysis Coceptuatilisation

It is clear that addressing this issue has the potential to become a step towards improving the quality of UHI assessment and making it more attractive for policymakers to use robust models instead of universal, locally ineffective solutions. Therefore, the focus problem of the owner could be stated as follows:

The current data-driven models for Urban Heat Island assessment fail to account for the spatial relationships of the features, which leads to suboptimal performance of the model.

1.2 Research Methodology

The present section focuses on designing the research to address the research problem formulated in the previous section. Firstly, the research objective is formulated and translated into a comprehensive list of research questions needed to reach this objective. Secondly, the research strategy describes how the research process is designed, followed by the outline of expected data collection and analysis. Lastly, an appropriate research schedule is developed to assign time-frames to different stages of the research strategy.

1.2.1 Research Objective

To be in line with the needs of the problem owner and cover the focus dimension of the challenge, the research objective could be formulated as follows:

To enhance the current data-driven UHI assessment methods by developing a systematic pipeline for creating a Machine Learning model that accounts for the spatial interdependency of temperatures.

1.2.2 Research Questions

To reach the presented research objective, several consecutive research questions are formulated. These questions follow a three-step reasoning: first, exploring potential modelling strategies; second, applying insights to improve the UHI model; and finally, evaluating the effectiveness of the proposed enhancements. Throughout this section, references to the "current UHI assessment model" or "baseline model" refer to the work of the problem owner, Pena Acosta [3].

It is evident from the problem statement and objective that current UHI assessment models do not adequately capture spatial interdependencies between features. However, it remains unclear how these spatial dependencies can be effectively modelled with the available data to enhance predictive performance.

This research assumes two potential directions for improvement to incorporate spatial dependencies: the use of more advanced Machine Learning algorithms, as recommended by Pena Acosta [3], and the adaptation of conventional models, such as Random Forest. In both cases, it is expected that improvements link to the development of appropriate feature engineering techniques. Feature engineering, in this context, refers to the process of transforming raw spatial and non-spatial data into meaningful inputs that reflect spatial relationships and are compatible with supervised learning.

To investigate these aspects, the first research question and its sub-questions are formulated as follows:

1. How can machine learning techniques be adapted to effectively model spatial dependencies of temperatures in Urban Heat Island intensity?

- **1.1. How can spatial feature engineering be designed reflect spatial dependencies of temperatures in urban street networks?**
- **1.2. Which machine learning algorithms are most suitable for capturing the interplay of spatial dependencies of temperatures effectively?**

Once suitable approaches for modelling spatial dependencies are identified, the next step is to evaluate whether these enhancements actually improve the model's effectiveness. This leads to the second research question, which focuses on assessing the value added by incorporating spatial relationships into the UHI modelling process.

The evaluation is grounded in a comparison with existing results, specifically those of Pena Acosta [3], whose Random Forest model serves as a baseline due to its strong performance in earlier assessments. By using this baseline results, it becomes possible to quantify the added value of the proposed enhancements.

This question aims to test performance changes from two complementary perspectives. Firstly, it looks at how well the improved model performs at different spatial scales. For instance, whether it yields stronger predictions at the street, neighborhood, or city level. This is important to understand whether the spatial enhancements are equally effective differeny spatial scales.

Secondly, since the final model is developed as a general pipeline, it is important to assess its applicability to other cities. Urban typologies including differences in environmnetal, urban and socio-economic characteristic can vary widely between contexts, which may affect model performance. Therefore, the seconds sub-question investigates whether the successful application of a pipeline observed in a trained setting are valid for unseen data. In other words, what is the performance of the pipeline in terms of its transerability to other cities.

2. How does model performance change when incorporating spatial dependencies between features using the proposed method?

- **2.1. How do the proposed enhancement methods affect UHI model performance across different spatial scales?**
- **2.2. How does performance of the proposed pipeline vary across different urban typologies?**

By answering these research questions, the research objective is met. The first question lays the foundation by identifying mechanisms for improving spatial representation in UHI modelling, while the second focuses on evaluating whether these improvements result in tangible gains in performance and spatial interpretability.

1.2.3 Research Strategy

To answer the research questions formulated above, a structured research strategy is designed. As in the majority of construction management challenges, the current paper follows a regulative cycle of applied research. This strategy is divided into three consecutive phases: (I) Literature Review and Theoretical Foundation, (II) Pipeline Development, and (III) Model Evaluation and Transferability.

Phase I establishes the theoretical foundation of the research and addresses both sub-questions of Research Question 1. It involves an extensive literature study into Civil Engineering, Data Science, and cross-domain sources. The goal of this phase is to identify suitable machine learning models and feature engineering strategies for capturing spatial dependencies, compare their effectiveness, and identify key gaps. Together, the outcomes form a theoretical base for the pipeline development.

Phase II translates the theoretical insights into practice. It focuses on the development of a data-driven pipeline for UHI assessment that accounts for spatial feature interdependencies. In this phase, a new feature engineering method is proposed to address the limitations of current approaches, and is merged with the most promising modelling direction identified in Phase I. This results in the implementation of the modelling pipeline, which is further optimised using a Genetic Algorithm.

Phase III tests the effectiveness and robustness of the developed pipeline. In line with Research Question 2, the enhanced model is evaluated against the baseline results of Pena Acosta [3], using the same Apeldoorn satellite dataset to ensure comparability. The comparison focuses on two aspects: the predictive performance, assessed using the coefficient of determination (R^2), and the feature importance analysis. The latter allows for interpreting whether the inclusion of spatial dependencies results not only in a more accurate but also more explainable model/

Beyond this, in Phase III, the pipeline's performance is tested across multiple spatial scales to evaluate sensitivity, and applied to new datasets to examine its transferability to other urban contexts with varying typologies and data structures. The overview of the research strategy, including all key activities, input sources, and deliverables, is presented in Figure 3.

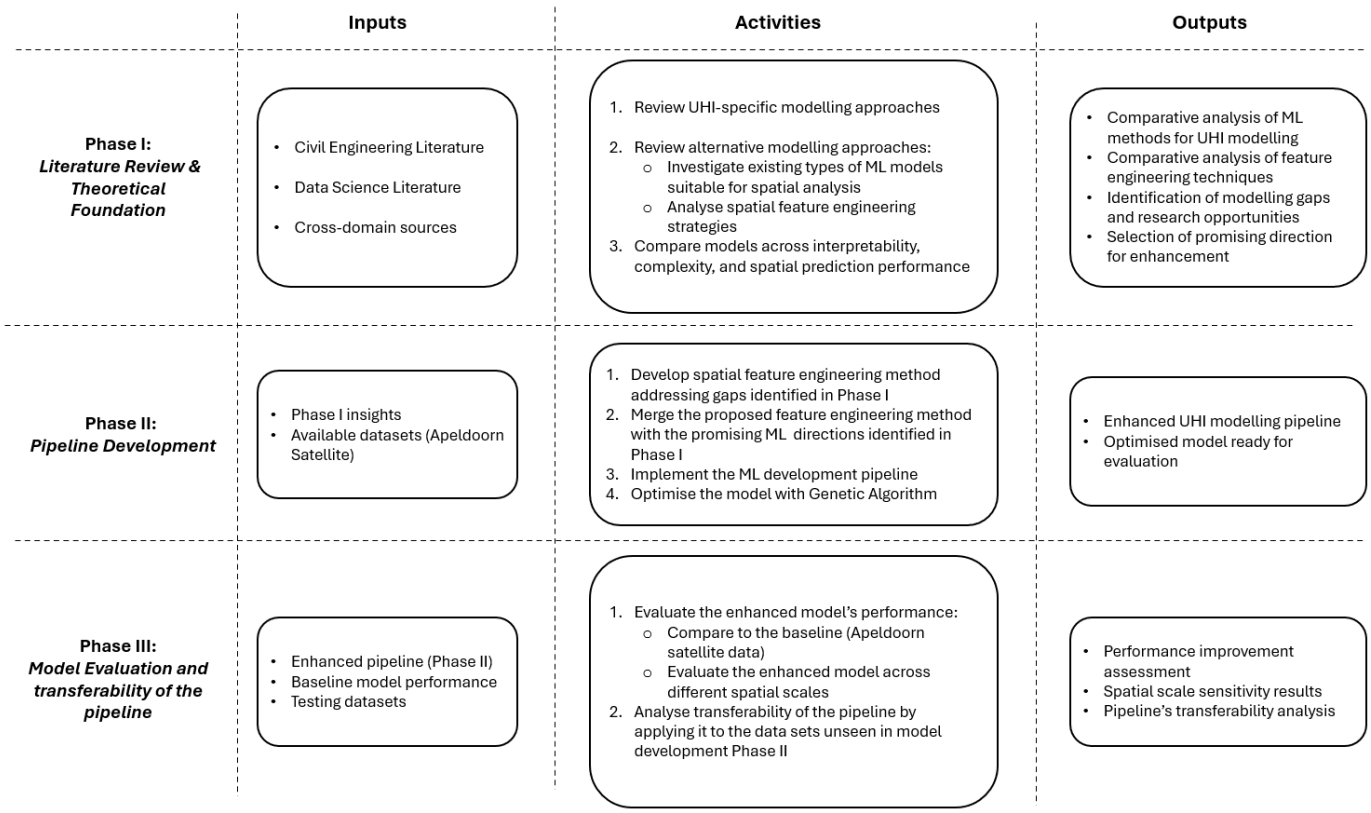


Figure 3: Research methodology framework

1.2.4 Data collection

The research strategy described above requires multiple sources of information. To make this information available in a structured and effective way, the data collection strategy needs to be developed.

In the first phase, the research uses several types of literature reviews that require different approaches. For the literature in the Civil Engineering domain, the bibliography list of Pena Acosta [3] should be used as a major source, since it collects an extensive amount of papers published throughout the 20th and 21st centuries. However, for the data science literature, there is no immediately known pool of relevant sources. Moreover, since the author of the designed thesis has a limited background in Computer Science, in-depth theoretical papers should be avoided in this domain. Instead, review articles and methodological surveys should be prioritised, which provide synthesised knowledge on machine learning techniques used to model spatial dependencies between features.

Furthermore, to explore alternative methods and feature engineering techniques suitable for capturing the spatial interplay between features, literature from adjacent fields such as environmental modelling, meteorology, and geography is investigated. To ensure transparency and reproducibility, keyword-based searches are conducted through academic search engines such as Scopus and Google Scholar. Search terms include phrases like "spatial feature dependence," "spatial autocorrelation machine learning," and "spatial feature engineering in environmental modelling." This approach enables the identification of relevant interdisciplinary literature in a rigorous and structured manner. All three literature review types and their corresponding objectives, methods, and tools used are summarised in Table 1.

Literature review field	**Objective**	**Methods**	**Tools Used**
Civil Engineering	Understand the nature of UHI data and implication for the modelling	Systematic review of existing literature	Pena Acosta [3] bibliography
Data Science	Identify ML methods for modelling dependent features	Review of synthesis articles and survey papers	Scopus, Google Scholar
Other Domains	Explore methods from other fields for capturing spatial dependencies	Cross-disciplinary keyword-based literature search	Scopus, Google Scholar

Table 1: Summary of Literature Review Types

1.2.5 Data analysis

The main dataset used for model training originates from the research of Pena Acosta [3] and has been used with the author's consent. It was collected from Landsat 7 satellite imagery for the city of Apeldoorn, the Netherlands, and includes 7166 samples, each representing a street segment. Each sample contains Land Surface Temperature (LST) measurements and a set of urban morphological and socio-economic features.

To evaluate the performance of the developed pipeline, three additional datasets were used:

1. Satellite-based datasets for the cities of Rotterdam
2. Satellite-based datasets for the cities of Montreal
3. A bike-collected dataset for Apeldoorn

While the Rotterdam and Montreal datasets are similar in nature to the Satellite-based Apeldoorn training set, the bike-collected set contains temperature measurements for both canopy and surface and it has a temporal resolution for 105 street segments forming a closed loop. Table 2 summarises the availability of features across all four datasets. The Apeldoorn satellite dataset was used for model training, while the Rotterdam, Montreal, and Apeldoorn bike datasets were used exclusively for testing.

Feature	Apeldoorn (Satellite)	Rotterdam (Satellite)	Montreal (Satellite)	Apeldoorn (Bike)
Building density	✓	✓	✓	✓
Vegetation density	✓	✓	✓	✓
Height–Width ratio	✓	✓	✓	✓
Width	✓	✓	✓	✓
Predominant facade	✓	✗	✗	✓
Predominant land use	✓	✓	✓	✓
Mean population	✓	✓	✓	✓
Mean building elevation	✓	✓	✓	✓
Maximum building height	✓	✓	✓	✓
Water density	✓	✓	✓	✓
Bike cover	✓	✗	✗	✓
Vehicle cover	✓	✗	✗	✓
Mean relative humidity	✗	✗	✗	✓
Reference relative humidity	✗	✗	✗	✓
Mean canopy temperature	✗	✗	✗	✓
Reference canopy temperature	✗	✗	✗	✓
Mean surface temperature	✗	✗	✗	✓
Reference surface temperature	✗	✗	✗	✓

Table 2: Feature availability across datasets

It can be seen that different simulaitons were enventually run with different number of features varying from 10 to 18. This misalignement was not specifically handled since the current paper focuses fully on the phase after the creation of the used data sets. Instead, all the simulation included as many meaningful features avaiilable in the provided data sets as possible.

In the bike-collected dataset, several additional features were available that introduced a temporal and physical dimension to the data. Depending on the prediction target, certain features were deliberately excluded to prevent recursive prediction:

- When predicting **Land Surface Temperature**, the features *Mean surface temperature* and *Reference surface temperature* were excluded.
- When predicting **Canopy Surface Temperature**, the features *Mean canopy temperature* and *Reference canopy temperature* were excluded.

Data preprocessing was performed using Python for automation and Excel for manual verification. This included handling missing values, duplicates, resolving formatting inconsistencies, and aligning features across datasets.

The model performance was assessed using the coefficient of determination (R^2) and feature importance rankings. All models were subject to hyperparameter tuning using a Genetic Algorithm (GA) to ensure reproducibility and performance optimisation. Both perfomance assessment and optimisation were performed similarly in Python environement.

2 Literature Review

Building on the gaps identified in [3], this review complements the Civil Engineering literature by exploring how spatial relationships are modelled in other scientific domains. Two main modelling streams emerge: spatial interpolation techniques and machine learning (ML) approaches. Among ML models, three categories are most commonly found in spatial applications: spatial regression models, ensemble-based learners, and deep learning architectures. The following sections discuss each category in more detail, with a comparative overview provided in Appendix A.

2.1 The Nature of UHI Data and Implications for Modelling

Urban Heat Island (UHI) research has evolved from early descriptive studies into a more quantitative and modelling-oriented field that aims to assess, predict, and ultimately mitigate the adverse impacts of urban heat. This progression is evident in the work of Pena Acosta [3], who offers an extensive summary of UHI modelling approaches within the Civil Engineering domain.

The literature commonly distinguishes between two types of UHI: Surface UHI (SUHI) and Canopy UHI (CUHI). SUHI is derived from satellite-based Land Surface Temperature, while CUHI is based on near-surface air temperature measurements. These types represent different layers of the urban atmosphere and are driven by distinct physical processes. Specifically, SUHI is influenced more by surface material properties and solar radiation, whereas CUHI is affected by air circulation, vegetation, and microclimatic interactions within the urban canopy layer [3]. The thesis also explains that these UHI types differ not only in physical behavior but also in their temporal and spatial resolutions. SUHI typically provides high spatial granularity but limited temporal coverage due to satellite overpass constraints, whereas CUHI offers continuous temporal data with lower spatial coverage. This mismatch can lead to divergent UHI patterns, posing a challenge for models aiming to capture both effects simultaneously [3].

Initially, modelling efforts relied heavily on physics-based simulation tools such as Envi-met, WRF-UCM, and EnergyPlus. These tools are capable of detailed microclimate simulations but are also computationally demanding and require extensive input data. As a result, the field has seen a shift toward data-driven approaches. More recent studies increasingly adopt machine learning techniques such as Random Forests and Neural Networks for classification and regression tasks related to UHI, especially in large-scale or data-scarce contexts [3]. These models are praised for their flexibility, scalability, and reduced data requirements. For example, Bilang et al. [8] used the WRF model to simulate UHI in Metro Manila under varying land-use scenarios, while others have applied Random Forests to satellite-derived datasets to estimate UHI intensity across diverse geographies [9].

However, despite their increasing popularity, data-driven models like RF and NN typically treat each spatial unit independently. They often lack mechanisms to capture spatial autocorrelation—a well-established factor in urban microclimate behavior. This limitation becomes especially critical when analyzing UHI at high spatial resolutions, such as the street-segment level, where spatial interdependencies can significantly influence local thermal patterns [3]. While some studies attempt to include spatial dependencies via statistical models or feature engineering techniques, these approaches often rely on strong assumptions, such as stationarity, or make use of rigid spatial structures that may not generalize well.

To overcome these challenges, newer approaches like spatial ensemble models and graph-based deep learning methods have been proposed. These frameworks offer greater flexibility for capturing spatial structure without relying on pre-defined assumptions. Yet, the field still faces major obstacles: particularly around the generalisability and calibration of such models across cities with different climates and morphologies.

Pena Acosta's thesis makes a significant contribution by proposing a modelling pipeline that integrates spatial dependencies directly at the street-segment level. According to the author, accurate modelling requires careful consideration of spatial granularity, consistent reference point selection, and alignment of measurement scales. Reference temperature selection, in particular, is highlighted as a critical source of variability across studies. The reference zone, used to calculate indicators such as $\Delta T = T_{\text{urban}} - T_{\text{rural}}$, is often defined inconsistently, making comparisons between cities or time periods difficult. Therefore, the thesis advocates for clearer methodological reporting and the potential development of standardized reference definitions [3].

Another major challenge addressed in the thesis is the issue of model transferability. Due to differences in urban morphology, local climate zones, and data availability, models trained on one city often perform poorly when applied to another. This is demonstrated with experimental results showing significant drops in predictive performance across cities, reinforcing the need for context-specific calibration.

Finally, the thesis highlights the importance of explainability in modelling. It makes use of explainable AI tools such as SHAP (SHapley Additive exPlanations) to interpret feature importance in both SUHI and CUHI

models. Comparative SHAP analyses show that the importance of features such as vegetation density and relative humidity may differ depending on the UHI type, spatial unit, and temporal context considered [3]. This underscores the conditional nature of feature relevance and the need for nuanced model interpretation.

Given these modelling challenges ranging from asynchronous SUHI/CUHI behavior and inconsistent reference temperature definitions to spatial heterogeneity and limited transferability, it becomes crucial to continue developing improved modelling approaches. The current paper builds on the above foundation by exploring methods that explicitly incorporate spatial relationships, including both classical and modern techniques. The following sections evaluate these methods in depth.

2.2 Deterministic Spatial Interpolation

In spatial datasets, nearby observations often exhibit similar values — a phenomenon known as spatial autocorrelation. Capturing these spatial relationships is essential for applications ranging from weather forecasting and environmental modelling to urban planning and public health. Spatial interpolation methods aim to estimate values at unmeasured locations based on surrounding observations, exploiting this dependency structure. In the context of UHI modelling, ignoring spatial relationships may lead to oversimplified predictions and loss of contextual information that is critical in urban environments.

One of the first algorithms explicitly designed to model spatial relationships in GIS data is deterministic spatial interpolation techniques that date back to the beginning of the 20th century and mainly target natural and socio-economic variables. [10] There are two most common non-regression methods that attempt to represent data in a surface described by a mathematical function, that is later used to predict a value of a variable at an unknown location using interpolation:

1. **Nearest neighbour (NN)**:

   NN is a straightforward method that partitions data into so-called "Thiessen polygons" that are defined around the locations with known features. For the prediction, the algorithms identify in which polygon the prediction location falls and assign the feature value of the location that forms this polygon. In other words, the NN method assigns the feature value of the nearest neighbour to a prediction location. [11]

2. **Inverse Distance Weighting (IDW)**:

   IDW is the first model that takes into account the influence of multiple locations on the observation. For this, IDW finds a weighted average of the feature values of the n nearest neighbours, which is often set to the number of observations in the set, as can be seen in Equation 1. [12] Moreover, IDW defines an important notation of the weight of the importance of the surrounding locations. As can be seen in Equation 2, these weights are inversely related to distance and acknowledge greater influence of the closer points: [10]

$$\hat{z}(s_0) = \sum_{i=1}^{n} w_i \cdot z(s_i) \tag{1}$$

$$w_i = \frac{d_i^{-p}}{\sum_{j=1}^{n} d_j^{-p}} \tag{2}$$

   - $\hat{z}(s_0)$ is the predicted value at prediction location s_0.
   - w_i is the weight assigned to observation $z(s_i)$ at location s_i.
   - d_i is the Euclidean distance between locations s_0 and s_i.
   - n is the number of nearest observations
   - p is an exponent that controls the influence of distance; larger values of p increase the influence of nearer points.

In spite of their notable simplicity, deterministic spatial interpolatio methods such as like IDW have shown to have a poor performance in urban environments, which is likely caused by a complex and fragmented nature of cities. For example, Holobaca et al. [13] found that IDW-based UHI interpolation in Cluj-Napoca failed to reflect intra-urban temperature variability, especially in mixed land-use areas. It was also suggested by the author that more modern hybrid approaches such as regression kriging or ensemble models should be used due to their potential to better capture both global and local spatial patterns.

While deterministic interpolation methods are easy to implement, they lack explanatory power and perform poorly in heterogeneous urban environments, which is the focus of the current work. in an attempt to better understand and incorporate spatial dependencies, regression-based approaches from spatial econometrics have gained popularity. The next section explores how models could be suited for studying how urban form and socio-economic variables influence UHI intensity.

2.3 Spatial Regression Models

Unlike Spatial Interpolation models that focus purely on predicting values based on surface functions, spatial econometrics focuses on understanding relationships and statistical dependencies in data. Spatial econometrics is a domain of science that is traditionally used to handle data with spatial dependence and spatial heterogeneity. This subfield of econometrics typically uses specialized regression techniques such as spatial lag models, spatial error models, and geographically weighted regression to capture how data observations in geographic space influence each other. [14]

Spatial Regression Models (SRM) are extensions of classical Linear Regressions models that are designed to capture autocorrelation, where autocorrelation is defined as a "property of data that arises whenever there is a spatial pattern in the values located on a map, as opposed to a random pattern that indicates no spatial autocorrelation."[15] Spatial Regression Models are primarily used in fields such as climatology, agriculture, ecology, public health, and urban economics since these domains need to process multivariate and spatially and temporally correlated data.[15] These models are designed to explicitly handle spatial relationships in data by incorporating spatial dependency structures into a regression model. As can be seen in Equation 3 (LR) and Equation 4 (SLR), this is achieved through modifications in the error terms or predictor variables to account for autocorrelation between locations.

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \tag{3}$$

Where:

- $\mathbf{Y}$ is an $n \times 1$ matrix (vector) of the dependent variable in a dataset with n data points.
- $\mathbf{X}$ represents the $n \times p$ matrix of independent variables, where p is the number of predictors.
- $\boldsymbol{\beta}$ is the $(p+1) \times 1$ vector of parameters to be estimated, including an intercept term.
- $\boldsymbol{\epsilon}$ is the $n \times 1$ random error matrix, assumed to be normally distributed.

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\Lambda}\mathbf{L} \tag{4}$$

Where:

- $\boldsymbol{\Lambda}$ is an $n \times 1$ error matrix, indicating the presence of spatial autocorrelation.
- $\mathbf{L}$ is an $n \times n$ spatial filter matrix, which represents the spatial structure.

Based on the studied literature, 3 main branches of modelling of the above-mentioned spatial structure can be identified: Spatial Lag Models (SLM), Spatial Error Models (SEM) and Geographic Regression Models (GRM).

2.3.1 Spatial Lag Models

A spatial lag is a phenomenon where "the value of a variable at a specific location is influenced by the values of that same variable at neighbouring locations". [16] In the context of the current thesis, this variable would be the Land Surface Temperature of all the street segments. Spatial Lag Models on top of the dependent variable incorporate a spatially lagged version of this variable into the regression equation, which captures direct spatial spillover effects. [14]

In terms of specifying the lag term in modelling spatial dependence with SLM, there are 3 major model types [17]:

1. **Spatial Autoregressive Model (SAR)**:

   SAR models are limited to modelling only the spatially lagged version of the dependent variable [14]. These models are particularly effective in capturing scenarios where the outcome variable in one area depends directly on neighbouring areas.

2. **Spatially Lagged X (SLX)**:

   The SLX model extends the spatial lag to include the independent variables, allowing it to account for spatial spillover effects [18]. Although not directly intuitive, for UHI assessment such a SLX could provide insights into the possible influence of local characteristics such as land use, vegetation cover, and building types on the adjacent areas.

3. **Spatial Durbin Model (SDM)**:

   The SDM extends is generalised to nest many of the spatial regression models [18]. SDM includes spatial lags of both dependent and independent variables, and hence it is more comprehensive for capturing the indirect influence of neighbouring locations on multiple variables [18].

The SAR, SLX and SDM models are mathematically described by the Equations 5, 6 and 7 respectively. [17]

$$y_i = \rho \sum_{j=1}^{N} w_{ij} y_j + \sum_{k=1}^{K} \beta_k x_{k,i} + \epsilon_i \tag{5}$$

$$y_i = \sum_{k=1}^{K} \delta_k \sum_{j=1}^{N} w_{ij} x_{k,j} + \sum_{k=1}^{K} \beta_k x_{k,i} + \epsilon_i \tag{6}$$

$$y_i = \rho \sum_{j=1}^{N} w_{ij} y_j + \sum_{k=1}^{K} \beta_k x_{k,i} + \sum_{k=1}^{K} \delta_k \sum_{j=1}^{N} w_{ij} x_{k,j} + \epsilon_i \tag{7}$$

Where:

- y_i is the dependent variable for the i-th observation.
- ρ is the spatial autoregressive coefficient.
- w_{ij} is the element at the i-th row and j-th column of the spatial weight matrix $\mathbf{W}$ of size $N \times N$, indicating the spatial relationship between observations i and j.
- y_j is the value of the dependent variable for the j-th observation.
- β_k are the coefficients for the independent variables.
- $x_{k,i}$ is the k-th independent variable for the i-th observation, out of K independent variables.
- δ_k is the spatial parameter for the k-th independent variable x_k.
- $\epsilon_i \sim N(0, \sigma^2)$ is the error term for the i-th observation, assumed to be normally distributed.

2.3.2 Spatial Error Models (SEM)

Unlike the Spatial Lag models that capture spatial correlation in the predictor or outcome variables, the Spatial Error Model (SEM) handles spatial dependency by capturing autocorrelation in the error terms. In SEM the residuals $\boldsymbol{\epsilon}$ (see Equation 8 and 9) are spatially correlated due to some unobserved spatial factors influencing the outcome. This makes SEM particularly useful when it is uncertain which spatial factors play a role in determining the outcome.

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \tag{8}$$

Where:

$$\boldsymbol{\epsilon} = \lambda \mathbf{W} \boldsymbol{\epsilon} + \boldsymbol{\nu} \tag{9}$$

Where:

- $\boldsymbol{\epsilon}$ is the $n \times 1$ vector of spatially correlated errors.
- λ is the spatial autoregressive coefficient for the error term, indicating the degree of spatial autocorrelation.
- $\mathbf{W}$ is the $n \times n$ spatial weights matrix, encoding the spatial relationships between observations.
- $\boldsymbol{\nu}$ is the $n \times 1$ vector of independent and identically distributed (i.i.d.) error terms.

In the context of UHI assessment, many potentially affecting factors may not be directly measured but can still influence the results. However, the current research line aims to establish a connection between the known set of urban and socio-economic features and the prediction of temperature differential. Therefore, it can be concluded that error-based model might not offer the desired insight into importance of the predictors and hence would not be a viable choice for modelling UHI phenomenon prediction.

2.3.3 Geographic Regression Models

Geographic Regression Models (GRM) represent a significant extension of traditional regression algorithms by allowing model coefficients to vary across space rather than assuming they are constant globally. This makes GRMs particularly effective for modeling spatial non-stationarity, i.e., when the relationship between predictors and the outcome varies by location [19]. Note that non-stationarity implied above is purely geographical. In other words, GRMs are not time-series models and do not require temporal data — their power lies in capturing geographically varying relationships within cross-sectional spatial data. This feature makes GRMs especially useful in UHI assessment, where the influence of variables like vegetation cover or impervious surfaces can differ across urban contexts. For instance, Zhao et al. [20] applied Geographically Weighted Regression (GWR) across several Texan cities, uncovering that the cooling effect of vegetation was significantly stronger in compact urban cores than in suburban areas with more dispersed development. There 2 types of Geographic Regression Models depending on how they account for spatial scales:

1. **Geographically Weighted Regression (GWR)**:

   GWR uses a single bandwidth for all variables, which assumes that the spatial influence of all predictors is consistent across the study area (see Equation 10). [19].

2. **Multiscale GWR (MGWR)**:

   MGWR extends GWR by allowing each variable to have a different bandwidth and hence capture dependencies between variables on different scales (see Equation 11). [21].

$$y_i = \mathbf{X}_i \boldsymbol{\beta}(\mathbf{u}_i, \mathbf{v}_i) + \epsilon_i \tag{10}$$

$$y_i = \sum_{k=1}^{K} \beta_k(\mathbf{u}_i, \mathbf{v}_i) x_{k,i} + \epsilon_i \tag{11}$$

Where $\beta_k(\mathbf{u}_i, \mathbf{v}_i)$ represents the location-specific coefficient for the i-th (or k-th) predictor, which can vary for each observation depending on the geographic coordinates $(\mathbf{u}_i, \mathbf{v}_i)$.

In the context of the UHI assessment, the qualities of GRM might be particularly useful since the effect of factors like population density, vegetation cover, and building materials may vary across different parts of a city. For example, greenery in the city centre might have a higher influence on the resultant temperature that is typically higher in this area compared to the suburbs where greenery might not have as strong an impact on an already lower temperature. This can help identify areas where certain factors have a stronger or weaker influence on urban temperatures, and hence provide more tailored advice for the urban development.

In general, applications of Spatial Regression Models to UHI problems are well documented. For example, Yin et al. [22] used a Spatial Error Model to investigate UHI in Wuhan, which uncovered latent spatial dependencies tied to floor area ratio and sky view factor. Moverover, Zhao et al. [20] applied GWR across multiple Texas cities, which revealed that vegetation's cooling effect varies significantly depending on urban morphology and density. These studies highlight how spatial regression techniques can support UHI mitigation planning despite of the limitations mentioned above.

However, classical spatial regression models require prior assumptions about linearity and data distributions, which limits their adaptability to complex and nonlinear urban data. Ensemble machine learning methods and particularly bagging approaches like Random Forest, offer an alternative by flexibly capturing complex interactions. The following section explores these techniques and their diverse spatial adaptations.

2.4 Ensemble Models

Ensemble models improve the robustness and accuracy of predictions by combining multiple models. They do not inherently capture spatial relationships but can be adapted for spatial contexts by adding spatial features or combining them with spatial interpolation techniques. Note that many of the studies described below were

performed on the raster spatial data, which raises a question of how applicable these findings are for network structures, which is the nature of the UHI data sets.

One of the characteristics of the urban environments is anisotropy — heat transfer gets disrupted by built structures, surface materials, and airflow obstructions. This makes classical Euclidean distance-based assumptions problematic, especially when applied to irregular urban networks like street segments. Therefore, the more intricate learning of Ensemble Models shows a promissing direction for accomodating the anisotropy of the built environment.

2.4.1 Bagging Models

Bagging is an ensemble Machine Learning method based on the Decision Trees or Classification And Regression Trees (CART). Although these models are known for a low bias and high explainability, they are generally considered to be prone to overfitting and have low prediction accuracy. [23] To overcome these challenges, bagging uses repeated sampling to generate the average of the predictions from all weak learners [24]. This reduces prediction error variance and makes the model more stable and accurate. [10]

As an alternative to bagging models, boosting models offers a sequential ensemble technique where each new model tries to correct the errors made by the previous models. However, due to novelty of application of these models in Civil Engineering, limited research and time available, the current paper does not further discuss boosting algorithms and focuses on bagging-based approaches. The bagging model dominating in the research is the Random Forest (RF) and its multiple variations:

1. **Random Forest (RF)**:

   The ordinary Random Forest (RF) is the most widely used bagging ML model that combines multiple decision trees in its prediction. [25] Moreover, it introduces random feature selection at each step during the creation of each CART to overcome possible issues with bootstrapped samples that may still be correlated due to the presence of the dominant features. [26] The RF prediction $\hat{z}(s_0)$ can be expressed as shown in Equation 12:

   $$\hat{z}(s_0) = f(x_1(s_0), x_2(s_0), \ldots, x_m(s_0)) \tag{12}$$

   Where:

   - s_0 represents the location of interest.
   - $x_i(s_0)$ $(i = 1, \ldots, m)$ are covariates at location s_0.

   This model has become one of the most frequently used data-driven tools in UHI studies thanks to its ability to handle nonlinear relationships as well as complex variable interactions [9]. In their global meta-analysis, Chen et al. [9] found RF to be the most consistently applied ML model in urban climate research, which is particularly true in the cases with diverse features such as NDVI (Normalized difference vegetation index), building height or land use. RF is often preferred when interpretability is a concern, as it provides insights into feature importance.

   Nevertheless, spatial autocorrelation between features is not directly included in the ordinary RF algorithm, except for the effect of indirect spatial correlation in features (covariates).[10] Therefore, there have been developed multiple modifications of the Random Forest to account for the autocorrelation in spatial data. One of the most straightforward modifications is including the latitude and longitude of the location as features in the prediction.[27] Although this method does not directly model spatial relation of different locations, this can enhance the prediction of the model and could be a prompt way to enhance the UHI assessment of Pena Acosta [3].

2. **Random Forest for spatial prediction (RFsp)**:

   RFsp is a straightforward adaptation of the ordinary RF model that incorporates spatial dependencies in the data by introducing the buffer distance maps as features for all the observations in the training set. Buffer distance maps are obtained by calculating Euclidean distances from all locations that are used as predictors of the observation location. Typically, in RFsp there are created as many buffer distance maps as there are observations. [28]

3. **Geographical Random Forest (GRF)**:

   Geographical Random Forest (GRF) is conceptually very similar to Geographically Weighted Regression covered in Section 2.3.3. GRF fits a separate local RF model for every location in the observation set

by using features of the neighbouring locations during training. For the prediction during deployment of the model, GRF then uses the local RF model closest to a given location. This allows GRF to account for spatial heterogeneity, making it better suited for datasets where relationships between variables differ across locations [29], which might be particularly useful in UHI assessment.

4. **Kriging and RFRK**:

   Kriging is a geostatistical interpolation technique. Unlike deterministic spatial interpolation methods, kriging is a probabilistic model that estimates unknown points by weighing the surrounding data points, taking into account the degree of variation between them. The degree of variation and its change with a distance in kriging are represented by a variogram model that measures the variance of differences between observed values as a function of the distance and direction between them. Kriging is considered to be a superior spatial interpolation due to its ability to take into account spatial correlation and quantify the interpolation error. [30]

   Multiple kriging methods have been developed. The most fundamental one is Ordinary Kriging (OK). Pure use of this method has a major limitation due to its assumption of second-order stationarity, which models the mean of the underlying random function as a constant. [10] In other words, OK assumes that covariance between two data points depends only on the distance separating them and not on their location, which might not be suitable for modelling complex spatially interdependent data such as in the UHI phenomenon.

   Regression Kriging (RK) and Kriging with External Drift (KED) are extensions of the OR that incorporate the influence of features at a target location. As can be seen in Equation 13, RK is a two-step approach that first models the general prediction trend with the Linear Regression and then applies the OR process to the residuals to capture the local spatial autocorrelation. KED is a similar algorithm that instead of splitting the data into a regression step followed by a residual kriging, models the trend (also known as drift) within the kriging equation itself using the external covariate information. It has been argued that in recent practice, KED and RK became the main geostatistical interpolation techniques. [31]

$$\hat{z}(s_0) = \mathbf{x}_0^T \cdot \hat{\beta} + \boldsymbol{\lambda}^T \cdot (\mathbf{z} - \mathbf{X} \cdot \hat{\beta}) \tag{13}$$

   Where:

   - $\hat{z}(s_0)$ is the predicted value at the prediction location s_0.
   - $\mathbf{x}_0$ is a vector of p predictors (or covariates) at the prediction location s_0.
   - $\hat{\beta}$ is the vector of estimated regression coefficients.
   - $\mathbf{X}$ is a matrix of predictors at the n sampling locations.
   - $\mathbf{z}$ is a vector of observations at n sampling locations.
   - $\boldsymbol{\lambda}$ is a vector of n kriging weights used to interpolate the residuals.

   Both RK and KED leverage the qualities of the deterministic regressions to directly incorporate covariates and capture the global trend in the data, which in the case of UHI assessment would be formed by covariates such as urban-morphological and socio-economic features. These methods also leverage the probabilistic nature of the Kriging process that on top of the fitted general trend focuses on modelling the local spatial variability of the residuals. Since the residuals are often smaller in magnitude compared to the original values and potentially exhibit local correlation patterns not captured by regression alone, RK and KED tend to remove systematic errors in the predictions. [32]

   By using the LR to predict the global trend, these models overcome the inability of OK to model the mean in the underlying random function dependent on the features. [10] However, one of the major limitations of the RK and KED processes is that they assume the general trend to be a linear combination of the features (covariates) of data [10], whereas spatially dependent data might exhibit more complex relationships between the features and a target variable.

   To allow for more complex modelling of a non-linear relationship between the prediction and the covariates, in some research, the regression model in Regression Kriging algorithm is replaced with a Random Forest, as can be seen in Equation 14. [33] The replacement of LR component with a RF yields a **Random Forest with Regressing Kriging (RF RK)** model, which is a hybrid approach proposed by several authors. [34, 33] It has been successfully tested in studies for modelling soil properties mapping and environmental data prediction, where similarly to UHI assessment, the spatial heterogeneity is significant. [34, 33]

$$\hat{z}(s_0) = \hat{f}_{RF}(\mathbf{x}(s_0)) + \hat{e}(s_0) \tag{14}$$

Where:

- $\hat{z}(s_0)$ is the predicted value at a new site s_0.
- $\hat{f}_{RF}(\mathbf{x}(s_0))$ is the Random Forest prediction with covariates $\mathbf{x}(s_0)$.
- $\hat{e}(s_0)$ is the kriging prediction of the residual.

RF RK is a promising model that could be a valuable enhancement of the UHI assessment due to its ability to model non-linear relationships between the features and a target variable combined with a geostatistical adjustment of the error term of the model that attempts to model spatial correlation effect. However, a major limitation of using a RF as a prediction algorithm for the global trend in data is that it is effective mainly within the range of covariate values exhibited by the training data. [33] Moreover, it has been shown that RF has a tendency to overfit data that have significant noise in them. [35] Finally, training and using RF is orders of magnitude more time-consuming compared to linear regression, especially when if the data set contains many covariates, which is the case of the UHI assessment. [33]

5. **Random Forest Spatial Interpolation (RFSI)**:

   Sekulić et al. argued that "none of the current approaches that aim to include geographical context in ML explicitly included the actual observations at the nearest locations of the prediction location as covariates." [10] In response to this, they developed a Random Forest Spatial Interpolation model that explicitly incorporates features from nearby observations as predictors as can be seen in Equation 15 opposed to a general Random Forest (Equation 12). [10]

$$\hat{z}(s_0) = f(x_1(s_0), \ldots, x_m(s_0), z(s_1), d_1, z(s_2), d_2, \ldots, z(s_n), d_n) \tag{15}$$

   Where:

   - s_i $(i = 1, \ldots, n)$ is the i-th nearest observation location from s_0.
   - $d_i = |s_i - s_0|$ represents the distance from the i-th nearest observation to location s_0.
   - $z(s_i)$ is the value at the i-th neighboring location.

   Although the authors don't explicitly acknowledge this, essentially RFSI is a Random Forest version of a spatial lag used in e.g. SAR, as explained in Section 2.3.1. Unlike kriging models, RFSI has no stationarity assumptions. Moreover, unlike RF RK, RFSI directly incorporates the features of neighbouring locations as covariates. These characteristics allow the RFSI to model non-linear relationships between features and the target variable spatially interdependent data [10], which makes it robust in modelling complex spatially interdependent data and a potential candidate for a significant enhancement of the UHI assessment.

Despite their flexibility and strong predictive performance, in the UHI context, ensemble models often rely on indirect spatial proxies such as geographic coordinates, engineered features from nearby observations, or distance buffers to account for spatial dependencies. While these techniques can enhance model accuracy, they might not fully capture how physical connectivity and built form shape temperature patterns in cities. This becomes especially relevant when working with network-based representations like street segments, where thermal interactions may follow the structure of the urban fabric rather than just geometric proximity. For instance, two streets separated by a building might be spatially close in Euclidean terms but thermally disconnected due to barriers to airflow and shading. These subtleties are difficult to represent with typical feature engineering approaches, pointing to the need for more spatially grounded modelling strategies that can directly incorporate the structure of urban space.

While ensemble models offer improved accuracy and interpretability, they still rely on engineered spatial features and may struggle to capture intricate spatial structures that are inherent in street network-based UHI data. Deep learning models such as graph-based neural networks, however, provide a powerful alternative by learning spatial dependencies directly from data topology. The next section reviews recent advances in deep learning for spatial modelling.

2.5 Deep Learning Models

Deep learning is a "subset of machine learning characterized by artificial neural networks with multiple layers, which are designed to learn complex representations of data through feature hierarchies." [36] One of the key

advantages of deep neural networks is that they can learn underlying nonlinear relationships between input and output data. Although there are many types and classifications of deep learning models, in the scope of the current literature review, three major branches are distinguished: Time-sliced, Temporal, and Spatiotemporal Deep Learning Models.

2.5.1 Time-sliced models

An advanced approach to handling time-sliced data is through Deep Learning methods, particularly neural networks, which offer a powerful means of capturing complex patterns in data. One of the most fundamental architectures is the Feedforward Neural Network (FNN), which processes input features through multiple hidden layers to learn hierarchical feature representations. FNNs are particularly effective in handling non-linear interactions between variables, making them a valuable alternative to traditional regression-based models [37]. However, standard FNNs lack inherent spatial dependency mechanisms, meaning they do not explicitly incorporate relationships between neighboring data points. Despite this limitation, FNNs may still offer strong predictive capabilities when applied to urban heat island modeling, especially when complemented with feature engineering techniques that introduce spatial characteristics into the input data. Their ability to model complex feature interactions makes them a viable comparison to other machine learning methods such as Random Forest.

A natural extension of FNNs is the Convolutional Neural Network (CNN), which is widely used for spatial data processing. CNNs introduce convolutional layers that enable local feature extraction, making them particularly effective for image-based tasks. However, conventional CNNs are designed to operate on regular grid structures, such as pixels in an image, and are not directly applicable to non-Euclidean spatial data such as road networks.

To address this limitation, Graph Convolutional Neural Networks (GCNNs) have been developed to generalize convolutional operations to graph-based data structures. GCNNs extend traditional CNNs by allowing the network to operate on irregular, graph-structured data, making them suitable for urban spatial analysis [38]. In GCNNs, a spatial network is represented as signals on the vertices of a graph, where information is aggregated from neighboring nodes [39]. GCNNs follow an aggregation scheme in which each node learns a deep representation of contextual information by propagating the characteristics of its neighbors.

While graph-based neural networks provide a natural extension for spatial data modeling, their primary applications in existing research have been in classification tasks, such as the categorization of building patterns [17]. Therefore, [17] proposed two novel methods: **Spatial Regression Graph Convolutional Neural Networks (SRGCNNs)** and their extension, **Geographically Weighted SRGCNNs**, which adapt neural network architectures for regression-based spatial modeling.

SRGCNNs, as defined by [17], adapt a traditional workflow of spatial regression models, with a key difference being the use of a graph structure instead of a spatial weights matrix. Moreover, unlike spatial regression models that require a complete matrix $\mathbf{X}$ and corresponding vector $\mathbf{y}$ at every observed point, the SRGCNN requires only a sample of the $\mathbf{y}$ observations. To achieve this, SRGCNNs use **semi-supervised training**, allowing them to consider all spatial units, even those without observed labels. This property might generate substantial value in modeling UHI phenomena, since in the application of the model in a real city, high-quality temperature measurements might not be available at all locations of interest, while other input GIS data such as street length is typically more accessible. The described differences in the application of these two paradigms are illustrated in Figure 4.

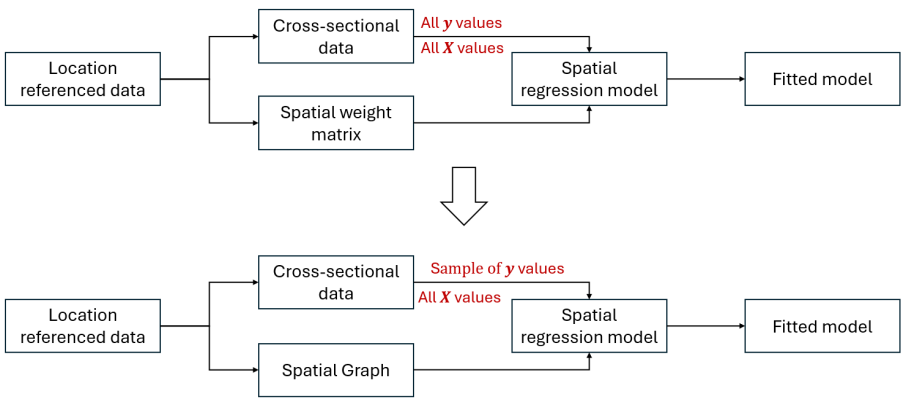


Figure 4: Comparison of SRM and SRGCNNs workflows

SRGCNNs model spatial multivariate distributions, similar to spatial regression models but using deep learning

methods, as seen in Equation 16.

$$\hat{y} = \mathbf{W}_{\mathrm{L}} \left(\sigma \left(\mathbf{W}_{\mathrm{L}} X \Theta^{(0)} \right) \Theta^{(1)} \right) \tag{16}$$

Where:

- $\hat{y}$ represents the predicted output for each spatial unit.
- X represents the input features for each spatial unit, similar to GCNN.
- $\mathbf{W}_{\mathrm{L}}$ is the normalized Laplacian matrix, capturing spatial connectivity.
- $\Theta^{(0)}$, $\Theta^{(1)}$ are trainable weight matrices for different layers.
- $\sigma(\cdot)$ is a non-linear activation function used to add complexity to the relationships modeled.

Similar to the concept of the geographic regression model described in Section 2.3.3, the Geographically Weighted SRGCNN explicitly considers spatial non-stationarity by incorporating different spatial weights matrices at each location to enable geographically weighted learning [17].

Graph-based deep learning starts gaining traction also in the UHI field. For example, Yu et al. [40] recently proposed a graph neural network architecture for estimating street-level heat intensity using spatial networks derived from OpenStreetMap and building datasets. Compared to CNNs and standard regression models, their GNN achieved lower RMSE and better generalisation across different urban typologies. These developments suggest that graph-based models could soon become standard tools for UHI modelling at the street scale.

2.5.2 Temporal models

Recurrent Neural Networks (RNNs) are a specific type of artificial neural networks that utilises recurrent connections that form loops, which allows the network to retain information across different time steps. This recurrent architecture makes RNNs particularly suitable for handling data that involves temporal sequences. [41] Although the spatial dimension of the urban data is not directly in the scope of the current thesis, the UHI phenomenon itself is a spatiotemporal process. Therefore, in this section and the following sections, an overview of ML models with a temporal component is provided as a guideline for future research. There are 2 major temporal models that appear in the research with an application to geographical data:

1. **Long Short-Term Memory Networks (LSTM)**:

   LSTMs are a type of Recurrent Neural Network (RNN) that is designed to capture long-term dependencies in sequential data such as natural language processing. This model utilises input-, output- and forget gates mechanisms to control information flow. [42] In spatial contexts, LSTMs can model sequences of spatial data points and effectively perform tasks such as land cover classification. [42]

2. **Gated Recurrent Unit (GRU)**:

   GRUs are a simplified variant of LSTMs that are more computationally efficient due to fewer number of gates. [43] GRUs are suitable for similar applications as LSTMs, including spatial data sequences. [**?**]

Interestingly, [37] reaches a conclusion that in most cases inclusion of prior spatial information might be not mandatory while using a neural network model for a spatiotemporal analysis. This raises the question of whether the above-mentioned temporal models could be used solely in spatiotemporal analysis. Nevertheless, due to a lack of research in this direction, it is further assumed that the model benefits from specifying prior information about the spatial dimension of data.

2.5.3 Spatial-temporal models

There is a direction of the research that approaches modelling spatiotemporal phenomena by leveraging the strengths of two major types of artificial neural networks covered in previous sections: Convolutional Neural Networks and Recurrent Neural Networks. These models are combined into a united architecture in various ways:

1. **Convolutional Long Short-Term Memory Networks (ConvLSTM)**:

   ConvLSTM is a hybrid model that extends LSTMs to have convolutional structures in their architecture. By using convolutions, the model can more explicitly account for the spatial relationships in predicting the precipitation nowcasting. [44]

2. **Graph Convolutional Recurrent Networks (GCRN)**:

   GCRN is also a hybrid model that utilises a combination of graph convolution for extracting spatial features from nodes and an RNN to capture temporal dynamics over time. [45] GCRN is conceptually similar to SRGCNN described in Section 2.5.1 with a main distinction being that the former also accounts for the temporal dimension by feeding the output of GCN layers into recurrent cells.

Moreover, there is a direction of research where Deep Learning models with a unique architecture are created to handle spatiotemporal data. For example, [46] proposes a novel **Spatio-Temporal Neural Networks (STNNs)** model that uses latent components to capture hidden dynamics of data. Similarly, this architecture has 2 components: one for encoding the spatiotemporal nature of the data into latent states, and one for decoding these latent states into predictions. [46] The latent state is a compressed and learned representation of both spatial and temporal structure in data simultaneously. This is the main distinction between STNNs and convulsion-based models that extract spatial information from the data before the temporal relationships are handled by RNN-like structures.

2.6 Comparison

With multiple modelling paradigms available, it can be argued that each has strengths and weaknesses. This section synthesises the reviewed approaches and compares them across key dimensions: spatial dependency, interpretability, scalability, and input data requirements.

Each of the researched methods can be partitioned into two sub methods: the Machine Learning algorithm itself that is applied to the data and the Feature engineering method that is used to shape data into the form processable and meaningful for the selected algorithm. Both parts are crucial for shaping a comprehensive model with an optimal performance. Therefore, both of these sub-dimensions are discussed separately below.

2.6.1 Machine Learning methods

It is debatable whether Deep Learning models are superior to ensemble models, since they typically require more training data and are generally less explainable. Although ensemble models do not inherently learn from spatial dependencies in data, there are multiple ways to incorporate these effects (see Section 2.4). Yet, DL models can capture the direct nature of complex spatiotemporal patterns and automatically extract features from data such as satellite imagery, which makes them particularly powerful for UHI assessment on a larger scale.

Moreover, some Deep Learning models such as SRGCNN have an advantage over the ensemble and Spatial Linear Regression methods since they do not require knowing the target variable values at all locations. Indeed, in SLR the use of spatial weights matrices to capture the spatial dependencies implies that it can only be defined among the locations where both independent and dependent variables are observed [17].As can be seen in, e.g., Equation 5, the spatial lag term $\rho \sum_{j=1}^{N} w_{ij} y_j$ requires the presence of the dependent variable around the predicted location. On top of the fact that this demands that the specification of spatial relationships can be defined only within a fully observed set, it can also cause overfitting [47] and is not suitable for transductive prediction under missing data [48]. Note that this disadvantage also holds for the described ensemble models.

Furthermore, both DL and ensemble methods have an advantage over the SLR models in terms of capturing the nature of dependencies in data. Namely, the applications of the model based on the principles of econometrics are limited by prior assumptions such as the linearity of the regression model and the normality of data distributions [49]. These models approach modelling spatial relationships through predetermined functional forms and linear combinations of variables, while these relationships might have an inherently non-linear nature [50].

Finally, in describing a complex system with both spatial and temporal relationships, the majority of the research uses time-sliced data, e.g. one satellite image of the surface, whereas UHI is a dynamic process that changes over time. The methods of econometrics used in SLR use straightforward fitting and hence might not be suitable to handle data with an inherent incompleteness of data when applied to these slices [51]. For example, some core dependencies might be underrepresented, while values with a low probability of occurrence might introduce significant bias in the data, which will also be learned by a linear model [52]. Of course, none of the models performs best on an incomplete or biased data. However, unlike SLR, Random Forest is generally equipped to deal with tools such as bootstrap sampling and random feature selection, while DL methods such as SRGCNN might not be less dependent on the values of the target variable.

Table 3 summarises the comparison between Spatial Regression Models, Ensemble Models, and Deep Learning Models for UHI assessment. SRMs offer simplicity and ease of interpretation, which is particularly useful in planning contexts [20] but are limited in their ability to model complex and non-linear spatial relationships. Ensemble models attempt to provide a balanced solution by providing robustness and some interpretability while

handling non-linearities effectively. While Deep Learning models are powerful in capturing both spatial and temporal dependencies, they require higher computational costs and offer limited transparency. Nevertheless, the is no solid research that would demonstrate an absolute domination in terms of performance of one of these models over the rest for predicting UHI phenomena. Therefore, all the 3 proposed directions in the ML techniques remain viable.

Model Type	Pros	Cons
Spatial Regression Models (SRM)	1. Simple and easy to interpret 2. Accessible to non-experts 3. Well-suited for linear relationships	1. Assumes linear relationships 2. Requires prior assumptions such as normality of data distributions 3. Limited applicability to complex geographically distributed phenomena 4. Use of spatial weights matrix limits flexibility and can lead to overfitting or issues with missing data
Ensemble Models	1. Handles non-linearity well 2. Provides feature importance 3. Robust to overfitting 4. Can incorporate spatial effects through feature engineering 5. No assumptions about data distribution	1. Computationally intensive 2. Requires hyperparameter tuning 3. Less intuitive than SRM 4. Does not inherently model spatial dependencies without feature engineering
Deep Learning Models (DL)	1. Captures spatiotemporal dependencies 2. Automatically extracts features 3. Can handle large-scale data effectively	1. Lacks transparency (black-box nature) 2. Requires large datasets and high computational resources 3. Difficult to interpret for urban planning purposes 4. Prone to overfitting if not regularized

Table 3: Comparison of ML Models for UHI Assessment

The choice of machine learning algorithm alone, however, is insufficient. The way spatial features are engineered plays a critical role in model performance. The following section categorises existing feature engineering approaches, identifies their limitations, and evaluates their fit for network-based UHI modelling.

2.6.2 Feature engineering methods

To gain a more complete understanding of the described machine learning approaches and to further decide which method is most perspective to investigate in the scope of the current paper, the overview of the feature engineering methods used in these models is presented in Table 4 and is elaborated below.

Spatial linear models have a simple tabular input structure along with the spatial weigh matrix. Spatial lag models such as SAR, SLX and SDM all use a lagged dependent variable as a covariate with a difference that SAR lags target variable of neighbours, SLX lags dependent variables of the neighbours and SDM lags both. All the lag models as well as spatial error model use spatial weight matrix to incorporate stronger influence of geographically closer covariates. [17] However, lagging "y-variable" makes SAR and SDM models not directly applicable for UHI modelling since this would lead to a recursive loop, where temperature prediction of a target location would be based on the temperature of direct neighbours, which fully covers all the variation in data and hence is not meaningful.

Geographical Weighted Regression is an extension of a regression that allows for the coefficients to vary per location. Therefore, for each location, a regression is fitted. However, this means that each location should also have the corresponding weight matrix reflecting influence of the neighbours. Therefore, although weight matrix is a core element in the GWR model, unlike in SLM models, no single matrix for the entire dataset can be defined. Instead, the corresponding matrixes are defined per location within the algorithm based on the geographical coordinates that are supplied as an input for training. Similarly, the weighted version of GWR calculates the location specific matrixes. This model also requires predictor bandwidth as an input for training, however it is automatically calculated using back-fitting algorithm. [21] Although GWR/MGWR might possess potentially advantageous qualities, it does not directly distinguish neighbours of the given segment as separate entities, unlike its non-geographical counterparts, which might reduce interpretability of results. Therefore, in the scope of the current research, it is decided not to continue in the direction of this method.

Similarly to linear models, ordinary random forest has a tabular input structure, but it does not directly model

spatial relationships in data unless certain feature engineered columns are appended to the data. An RFsp extension of RF require buffer distances map as the input to model spatial relationships. The maps are simply euclidean distances between the locations. Although, in principal, similar to the spatial weight matrixes, buffer maps are typically not normalised. Moreover, for large data sets, RFsp and yield to dimensionality inflation since the number of maps needed corresponds to the number of observations in data. [28] Therefore, this method is not suitable for the scope of the current paper. A predecessor of RFsp that was extensively compared to it is a RF Random Kriging model. RF RK combines RF predictions with residual kriging. Therefore, in principle, the RF RK needs the same input as an ordinary Random Forest. It does additionally need variogram parameters such as nugget, sill, and range to model spatial autocorrelation in the residuals. However, these are estimated and fit to the residual of the RF prediction within the model. Nevertheless, it has been shown that RF RK does not outperform simple non-statistical extensions of RF, while it also introduces rigid statistical assumptions and experiences reduction in performance on zero-inflated data sets. [28] Therefore, it has been decided not to continue further with RF RK in the scope of the current research.

Geographical Random Forest fits a local random forest model for each observation in the training set using the features of neighbours. In GRF, the neighbours are identified adaptive kernel by selecting the n-nearest observations or using fixed kernel that defines neighbours within a fixed radius or distance from the target location. In either case, the bandwidth parameter need to be specified as the input of the modes that would determine the spatial extent of the kernel. For mathematical operations with these kernels, the geographical coordinates are also included as explanatory variables as an input for GRF training. [29] Although GRF has a potential to capture the spatial relationships in the UHI data, fitting a new RF model at each location poses a significant challenge of computational expensiveness of training, which inflates manifolds during optimisation of model's hyperparameters. Therefore, it has been decided not to continue further with GRF in the scope of the current research.

RFSI is comparable in its core to Spatial Lag Models (SLMs) but has substantially different data structure. Similarly to SLMs, it requires features of neighbouring locations as an input for each target location. However, RFSI does not extract these features from the feature matrix within the algorithm and does not use spatial weight matrixes. Instead, RFSI requires the input to contain features of neighbouring locations and the distance to these locations in each row of a feature matrix used for training. [10] Nevertheless, like in the case of RFsp, the feature engineering needed for RFSI leads to dimensionality inflation for dense datasets with substantial number of neighbours. Therefore, it has been decided not to continue further with RFSI in the scope of the current research.

Similarly to an ordinary RF, a NN requires a simple tabular data that does not directly capture influence of adjacent streets, which can, however, be augmented with feature engineering. A convolutional counterpart of a NN, the SRGCNN is a graph-based model that requires a spatial graph as an input, where nodes and edges to represent individual spatial units and connection between them. In principle, the spatial graph can be defined directly from the spatial weight matrix. Nevertheless, graph is a separate structure and hence should be explicitly treated differently. Moreover, the SRGCNN model requires several training parameters, such as Training Mask that separates training nodes with y-values from training nodes without y-values in the semi-supervised process. Similarly to GWR, the weighted version of SRGCNN explicitly considers the spatial nonstationary by incorporating different spatial weights matrix and the corresponding parameters at each location. Similarly, the weighted SRGCNN incorporates this effect within the algorithm by introducing geographically weighted layer-wise graph convolutions. [17] Therefore, no additional data input structures are needed for this model.

ConvLSTM is a spatio-temporal model and hence it requires a spatial grid tensor for each time step. This tensor is an extension of a 2D feature matrix into the time dimension that represents the state of all spatial units at each time step.[44] Similarly, GCRN require observations of a signal at each time step, with features for each node.[45] STNN uses the tensor structure for storing and processing the feature data at each time step, while graph structures are leveraged for explicit modelling of spatial relationships. Therefore, both structures are needed as an input of the STNN model. [46] Since the dataset used in the current research does not have a temporal dimension, all the models requiring temporal and spatio-temporal feature engineering, are not suitable for the scope of the present paper.

* = Calculated/extracted within the algorithm ** = Inflated into the time dimension

2.6.3 Synthesis

Following the analysis above, except for dimensionality inflation and computational demand concerns, all the studied feature engineering methods have three challenges with respect to predicting the UHI phenomenon.

Firstly, the most suitable methods such as SLX or SRGCNN fit the models to the entire set immediately. This

	Features matrix	Spatial weight matrices	Features of neighbouring locations	Target variable at neighbouring locations	Buffer distance map	Geographical coodinates	Kernel bandwidth	Spatial Graphs	3D Spatial Grid Tensor	Dynamic adjacency matrix
SAR	X	X		X*						
SLX	X	X	X*							
SDM	X	X	X*	X*						
SEM	X	X								
GWR	X	X*	X*			X				
MGWR	X	X*	X*			X				
RF	X					X				
RFsp	X				X					
RFSI	X		X		X					
GRF	X		X*			X	X			
RF RK	X					X				
SRGCNN	X		X*					X		
Weighted SRGCNN	X		X*					X		
ConvLSTM									X	
GCRN	X**							X		X
STNNs								X	X	X

Table 4: Input data structure of ML algorithms

implies that it is not possible to study the importance of a spatial scale of predictor features, which might be an important factor in understanding urban interdependencies. Secondly, even though models such as RFSI or GRF essentially have suitable properties for modelling UHI phenomenon, their feature engineering methods introduce a significant dimensionality inflation issue when extended to a larger data set, which is the case in the present work.

Thirdly, all the feature engineering methods are largely developed for spatial non-network modelling purposes. This leads to an assumption that the importance of the neighbours' features decreases inversely proportional only to the Euclidean distances between the centroids of the target segment and the neighbours. However, UHI is a phenomenon specific to a city structure, which is, in essence, an interconnected network of street segments with natural barriers (such as buildings) between them. In reality, in the built environment, heat transfer and UHI effects depend not only on proximity but also on network structures, material properties, and airflow patterns. As a result, a physically distant parallel street may have a stronger influence on local temperatures than a directly connected segment. This nature of a modelled subject is not accounted for in the existing feature engineering methods and hence leads to several limitations. Namely, according to the inverse distance principle, the parallel is likely to have a larger influence on a given street segment than its direct neighbour since $d_1 < d_2$ (see Figure 5) even though there might be a significant natural or artificial barrier isolating two parallel streets.

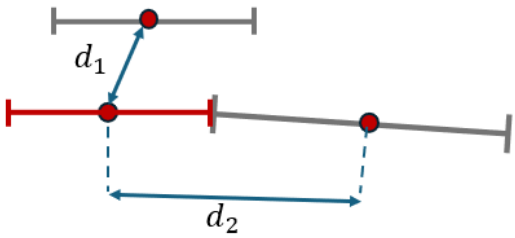


Figure 5: Centroid connectivity: parallel streets

Another potential limitation of the inverse-centroid-based-distance assumption pertains to the inclusion of longer streets. Similarly to the previous limitations, if a neighbouring street is long enough, it is going to be considered less influential compared to the parallel segments even though the longer street is a direct neighbour of a given segment.

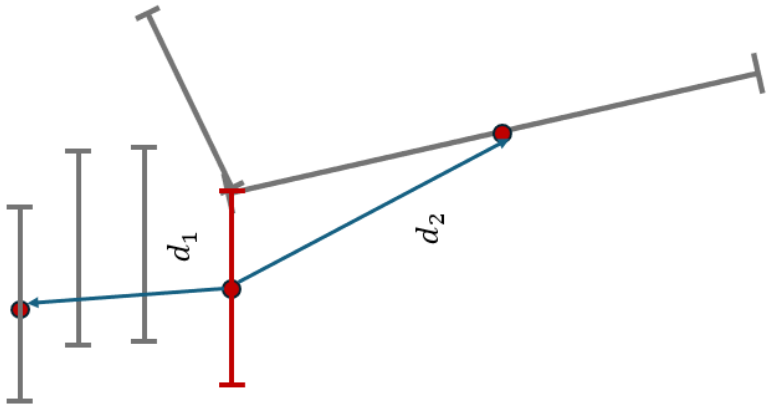


Figure 6: Centroid connectivity: parallel streets

Therefore, it can be concluded that a new data feature engineering method needs to be developed to address the three limitations mentioned above. Since, in principle, feature engineering can be decoupled from the training phase, the new data processing approach can be complemented by any one of three identified directions in Machine Learning: Linear Models, Ensemble Models and Deep Learning algorithms.

3 Proposed Method

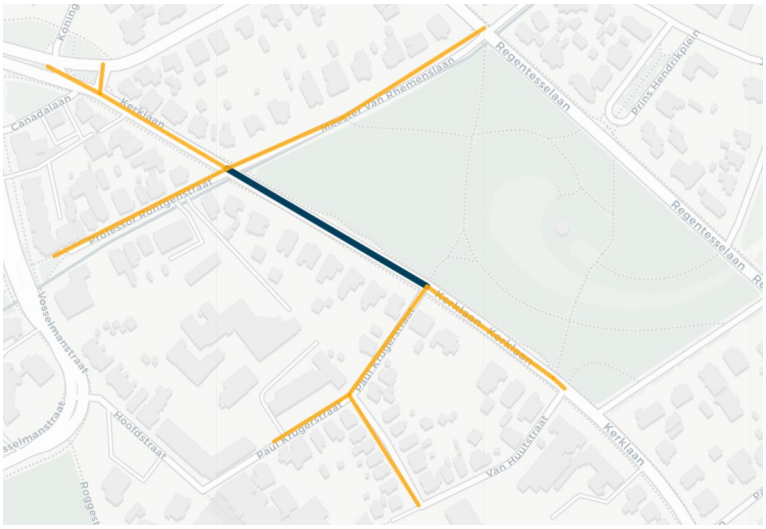
3.1 Core Enhancements

Firstly, to enable variation of a spatial scale in the model, the proposed method suggests including only a number of neighbours that fall within a certain threshold reflecting geographical proximity. By varying this threshold and hence the scale of neighbours' search, the model allows examining the scope of the meaningful contribution of neighbouring streets to a prediction.

Secondly, to tackle a dimensionality inflation challenge, the proposed method suggests using aggregation of the neighbours' features of the same type instead of including all of them as separate predictors in a training set. There are multiple ways to aggregate data, and hence several options are examined in the present work to explore the direction of best-performing feature aggregation methods.

Thirdly, to address the potential limitation of the inverse-distance proportionality of feature importance, an alternative node-based approach is suggested. Following the threshold concept introduced above, the neighbours can be defined as segments that are within a certain number n of intersections (later referred to as "nodes") that need to be passed between the potential neighbour and a given street segment. For example, if a threshold is 0, only the streets that are directly adjacent to a given segment will be selected (see Figure 7a), while for the threshold value 1, all the segments that require at most 1 intersection while travelling through the city network to a given segment will be selected (see Image 7b).

(a) Node-Based neighbour selection within 0 nodes



(b) Node-Based neighbour selection within 1 node

Figure 7: Comparison of Node-Based neighbour selection

Similarly to an inverse distance method, once the neighbours are identified, the feature importance would be inversely proportional to the number of intersections (nodes) that need to be travelled through the network from a given segment to a considered neighbour. This should prioritise features of the neighbours that are directly connected to a given street segment and decrease the feature importance not based on a network instead of

a simple geographical proximity. Since the intersections-based approach is a hypothetical enhancement of an inverse distance method, the latter needs to be used as a performance evaluation baseline. Therefore, the current research extends the proposed feature engineering method for both node (intersection) and radius (distance) based selection.

3.2 Model Variations

There are three components that create variation in the proposed method: neighbour selection, feature aggregation and model training. As explained in the section above, the neighbours can be selected either based on intersections (nodes) or distances (radius) thresholds. Once the neighbours are selected, to avoid the dimensionality problem, their features need to be aggregated. Since, there are multiple ways to perform aggregation, 4 of the following approaches are all separately tested:

1. **Separate Means**: The original segment features are preserved, and an additional set of features is created by averaging the attributes of all the identified neighbouring segments.
2. **United Means**: Each feature is replaced by the average of its own value and the mean of its neighbours. This aggregation can be thought of as an additional averaging of the feature groups from the Separate Means method.
3. **PCA Means**: This is analogous to the United means. However, instead of an arithmetic average, Principal Component Analysis (PCA) Transformation projection is applied to the segment and its neighbours to derive a lower-dimensional representation.
4. **Weighted Means**: The original segment features are preserved as in the Separate Means. However, neighbours contribute with weights based on network adjacency and distance instead of a simple mean, ensuring that closer and more connected segments exert a stronger influence. The comparison of Weighted Means with the Separate Means aggregation method is expected to give insights into the importance of weighting of features inversely to the spatial factor (nodes/distance) versus a simple averaging of the neighbours within a selected threshold.

Following the conclusion of the literature review, the core disadvantages of existing models can be mitigated by the proposed neighbour section and feature aggregation method. Hence, the proposed data processing could be further used for training using the 3 identified ML directions. However, there are two major concerns with the linear models. Firstly, it is highly unlikely that relationships between urban/socio-economic features and temperature gradient are linear. Secondly, and more importantly, the structure of linear models such as LR and SLX does not allow variation of a spatial scale due to its inherent structure, which cannot be directly resolved by the proposed method, unlike in the case of RF and NN.

Therefore, linear models are not chosen as a primary focus of the present research, although some of them are tested later for the sake of a baseline comparison with RF and NN-based models and confirmation of the hypothesis of nonlinearity of data dependencies.

To perform such a comparison, the standard SLX model that assumes all neighbouring segments contributing proportionally needs to be adjusted to enable variation of the spatial scale. Instead of aggregating all neighbouring observations, a spatial constraint was introduced where only neighbours within a certain radius R contribute. In contrast to the aggregation methods described above, which create new features through direct arithmetic or weighted averaging, the SLX (Spatial Lag of X) approach simply uses spatially lagged versions of existing features within training. These are not additional covariates but rather inputs aggregated within training to capture spatial spillover effects.

For each feature k, the SLX model defines a spatial lag term across its neighbours computed via inverse-distance weighting:

$$\tilde{x}_i^{(SLX)}(k) = \sum_{j \in \mathcal{N}(i,R)} \tilde{w}_{ij} \cdot x_j^{(k)} \tag{17}$$

With

$$\tilde{w}_{ij} = \frac{1/d_{ij}}{\sum_{j \in \mathcal{N}(i,R)} 1/d_{ij}} \tag{18}$$

The full lagged feature vector for segment i is then:

$$\vec{\mathbf{x}}_i^{(SLX)} = [\tilde{x}_i^{(SLX)}(1), \ldots, \tilde{x}_i^{(SLX)}(m)] \tag{19}$$

This lagged vector is incorporated into a linear regression model as:

$$y_i = \beta_0 + \vec{\boldsymbol{\beta}}^\top \vec{\mathbf{x}}_i + \vec{\boldsymbol{\delta}}^\top \vec{\mathbf{x}}_i^{(SLX)} + \varepsilon_i \tag{20}$$

Where:

- $\vec{\mathbf{x}}_i$ is the original feature vector of segment i.
- $\vec{\mathbf{x}}_i^{(SLX)}$ is the spatially lagged vector.
- $\vec{\boldsymbol{\beta}}$ and $\vec{\boldsymbol{\delta}}$ are coefficient vectors for the local and lagged features respectively.
- $\mathcal{N}(i, R)$ is the set of neighbors within radius R.
- $\tilde{w}_{ij}$ is the normalized inverse-distance weight.

Note that the indixes and notations above as well as in the follwonig sections of the present paper have the following nature:

- $i \in \{1, \ldots, N\}$ indexes the street segments.
- $k \in \{1, \ldots, m\}$ indexes the feature types.
- $\vec{\mathbf{x}}_i = [x_i^{(1)}, \ldots, x_i^{(m)}] \in \mathbb{R}^m$ denotes the feature vector of segment i.
- $\mathcal{N}(i)$ is the set of neighbors of segment i, determined via radius or node thresholds.

Unlike the Weighted Means method (Section 3.3.3), which embeds weighted neighbor features into the input dataset for nonlinear models, SLX incorporates spatial spillovers during model training. It maintains a clear separation between the original covariates and their lagged counterparts, enabling interpretation of direct vs. indirect (neighbor-driven) effects through $\vec{\boldsymbol{\beta}}$ and $\vec{\boldsymbol{\delta}}$ respectively.

This formulation ensures that only local neighbours contribute to the spatial lag and different radius values can be tested to analyse the model's sensitivity to spatial scale. The radius parameter R controls how local or global the spatial interactions are, allowing for scale sensitivity analysis.

The described above division in neighbour selection, feature aggregation and model training yields 16 variations of the model after exculsion of the linear models, as can be seen in Figure 8. Namely, all the researched ensemble models are based on the Random Forest (RF), while Deep Learning models are based on Fast Forward or Convolutional Neural Networks.

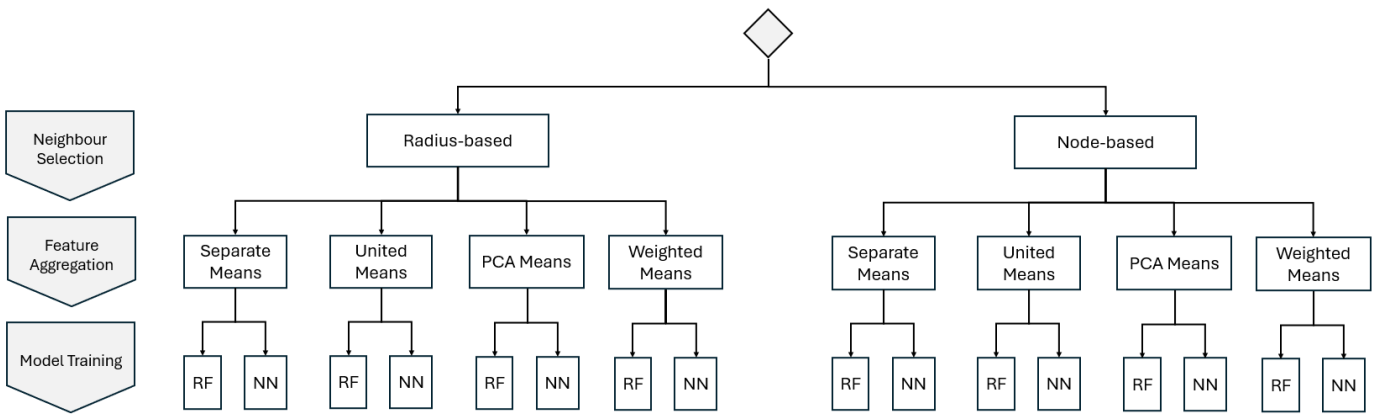


Figure 8: Model Variations

3.3 Neighbour Selection

The proposed intersection(node)-based neighbour selection necessitates calculating the number of intersections on the shortest possible path that connects two segments of the city. To realise such a calculation, a graph structure is used to model the street segments. Therefore, the selection of neighbours is performed using a graph-based representation of the urban road network and relies on two different distance metrics: Geographic Proximity (Haversine Distance) and Graph Proximity (Shortest Path Distance). These two metrics allow the definition of neighbourhoods on either physical distance or network connectivity, which covers both of the proposed neighbours selection methods.

3.3.1 Graph Construction

The road network is modelled as an undirected graph $G = (V, E)$ constructed based on the coordinates of endpoints of each street segment.

- V is the set of nodes, each representing an *intersection* or *road endpoint.*
- E is the set of edges, each representing a *road segment.*

Each edge $e_{ij} \in E$ connecting two nodes $v_i, v_j \in V$ is associated with a feature vector representung the features of segment e_{ij} (e.g., building density, vegetation density, width, etc.). These features are stored as edge attributes in the graph that get accessed later during the feature aggregation process.

Note that the node-based neighbour selection requires the existence of a path between the given segment and the rest of the graph. However, street data might be spatially fragmented, which is the case with the available data. Therefore, the isolated segments need to be removed, which was achieved by extracting the largest connected component from G, ensuring that only fully connected road segments are considered in the analysis.

3.3.2 Threshold Matrices

Both geographic and graph proximity characteristics are unique and stationary for a city network. Therefore, for the sake of computational efficiency, two matrices are pre-computed to define neighbour relationships:

The Haversine formula is used to compute the great-circle distance between the centroids of all road segments:

$$d_{ij}^{(H)} = 2R \arcsin\left(\sqrt{\sin^2\left(\frac{\phi_i - \phi_j}{2}\right) + \cos(\phi_i)\cos(\phi_j)\sin^2\left(\frac{\lambda_i - \lambda_j}{2}\right)}\right) \tag{21}$$

Where:

- R is the Earth's radius (6371 km),
- (ϕ_i, λ_i) are the latitude and longitude of segment i,
- (ϕ_j, λ_j) are the latitude and longitude of segment j.

This $N \times N$ matrix provides pairwise geographic distances between segments.

The network-based shortest path distance is computed using a graph search, Dijkstra's method. This method computes the number of edges in the shortest path between two segments. This calculation is equivalent to finding the number of nodes (intersections) under an assumption that "reaching" the target street segment is travelling to an edge that has a direct connection to one of its nodes.

$$d_{ij}^{(G)} = \min \sum_{(p,q) \in P_{ij}} w_{pq} \tag{22}$$

Where:

- P_{ij} is the set of paths between segments i and j,
- w_{pq} is the number of edges

This $N \times N$ matrix provides the shortest path distance in the road network.

3.3.3 Neighbour Selection

Once the threshold matrices are computed and the graph is constructed, the neighbours can be selected. Following the logic described earlier, for each road segment i, the set of neighbours $\mathcal{N}(i)$ is defined based on one of the two thresholds:

1. **Radius-Based Selection**: $\mathcal{N}(i) = \{j \mid d_{ij}^{(H)} \leq r\}$
2. **Node-Based Selection**: $\mathcal{N}(i) = \{j \mid d_{ij}^{(G)} \leq g\}$

Where:

- r is a geographic radius threshold (e.g., 100m, 500m, etc.).
- g is a graph distance threshold (e.g., 2, 4, 6 nearest segments).

3.4 Feature Aggregation

Once neighbours are selected, their features are aggregated into a new representation for each road segment. As described above, the four aggregation methods are used, which are applied to the features of the neighbours $j \in \mathcal{N}(i)$.

For the first three aggregation methods, feature aggregation is computed using basic arithmetic operations:

1. Separate Means For each feature k, the mean value is computeed across neighbors:

$$\vec{\mathbf{x}}_i^{(N,\text{sep})} = [x_i^{(N,\text{sep})}(1), \ldots, x_i^{(N,\text{sep})}(m)] \tag{23}$$

Where

$$x_i^{(N,\text{sep})}(k) = \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} x_j^{(k)} \tag{24}$$

This produces a new vector that summarizes the average characteristics of the neighboring segments with an output of:

$$(\vec{\mathbf{x}}_i, \vec{\mathbf{x}}_i^{(N,\text{sep})}) \in \mathbb{R}^{2m}$$

2. United Means The feature vector of the segment is averaged element-wise with the aggregated neighbor vector from the Separate Means method:

$$\vec{\mathbf{x}}_i^{(\text{uni})} = \frac{1}{2}(\vec{\mathbf{x}}_i + \vec{\mathbf{x}}_i^{(N,\text{sep})}) \tag{25}$$

This formulation blends local and contextual information into a single representative feature vector with an output of:

$$\vec{\mathbf{x}}_i^{(\text{uni})} \in \mathbb{R}^m$$

3. PCA Means For each feature k, form a local context vector that contains the value of the feature for the segment itself and for all its neighbors:

$$x_i^{(\text{pca})}(k) = \text{PCA}\left(\left[x_i^{(k)}\right] \cup \left\{x_j^{(k)} \mid j \in \mathcal{N}(i)\right\}\right) \tag{26}$$

This operation is then repeated for each feature type $k = 1, \ldots, m$, yielding a transformed vector where each component is a PCA projection of the local feature distribution:

$$\vec{\mathbf{x}}_i^{(\text{pca})} = [x_i^{(\text{pca})}(1), \ldots, x_i^{(\text{pca})}(m)] \tag{27}$$

The result is a dimensionally consistent vector that integrates spatial context via unsupervised learning with an output of:

$$\vec{\mathbf{x}}_i^{(\text{pca})} \in \mathbb{R}^m$$

4. Weighted Means Unlike the other methods, weighted means assign a varying influence to each neighbour, where the local weights are computed as a combination of geographic and graph-based proximity.

First, weights that blend proximity in terms of graph distance and Euclidean distance using local normalisation are defined. For each segment i and its neighbor $j \in \mathcal{N}(i)$:

$$w_{ij} = \alpha \left(1 - \frac{d_{ij}^{(G)}}{\max_{j \in \mathcal{N}(i)} d_{ij}^{(G)}}\right) + \beta \left(1 - \frac{d_{ij}^{(H)}}{\max_{j \in \mathcal{N}(i)} d_{ij}^{(H)}}\right) \tag{28}$$

where $d_{ij}^{(G)}$ is the graph (shortest-path) distance and $d_{ij}^{(H)}$ is the geographic (Euclidean) distance between segments i and j. Note that the normalisation is computed locally—$\max_{j \in \mathcal{N}(i)}$ is taken only over the neighbors of segment i.

Mext, the convex combination constraint is introduced to ensure that the weighted combination remains bounded and interpretable:

$$\alpha + \beta = 1 \tag{29}$$

Moreover, the resultant weights are normalised sum to one for each segment, which ensures that neighbor features contribute proportionally, avoiding artifacts from differing neighborhood sizes:

$$w_{ij}^{(\text{norm})} = \frac{w_{ij}}{\sum_{j \in \mathcal{N}(i)} w_{ij}} \tag{30}$$

Finally, these weights are applied to compute a weighted average for each feature k,

$$x_i^{(N,\text{w})}(k) = \sum_{j \in \mathcal{N}(i)} w_{ij}^{(\text{norm})} x_j^{(k)} \tag{31}$$

This results in the final neighbor-based feature vector:

$$\vec{\mathbf{x}}_i^{(N,\text{w})} = \left[x_i^{(N,\text{w})}(1), \ldots, x_i^{(N,\text{w})}(m) \right] \tag{32}$$

This output preserves the original segment vector and appends the locally normalized weighted contextual information, yielding:

$$(\vec{\mathbf{x}}_i, \vec{\mathbf{x}}_i^{(N,\text{w})}) \in \mathbb{R}^{2m}.$$

The four methods proposed above have different dimensionality in their data structure. If the dataset originally has $m = 12$ features, the Separate and Weighted Means aggregation methods contain $2m = 23$ features. Half of the variables cover features of the target segment, while the other half is formed by the selected neighbours. Conversely, the United Mean and PCA mean methods produce m features for training, where the features of the target segment and of neighbours get further aggregated. The latter is intended to test how the ML model performs under a varying number of predictors and what is the optimal extent of aggregation.

Note that in all the described aggregation methods, all the operations are performed on the features in a type-wise manner. For example, in the Separate Means aggregation, the mean is calculated across the rows of all the identified neighbours with the same feature type e.g. "Land Use" to create an aggregated "Neighbour Land Use" feature value. This yields $m = 12$ columns characterising the influence of the neighbours, which gets augmented with the $m = 12$ features of the original street segment, as can be seen in Figure 9.

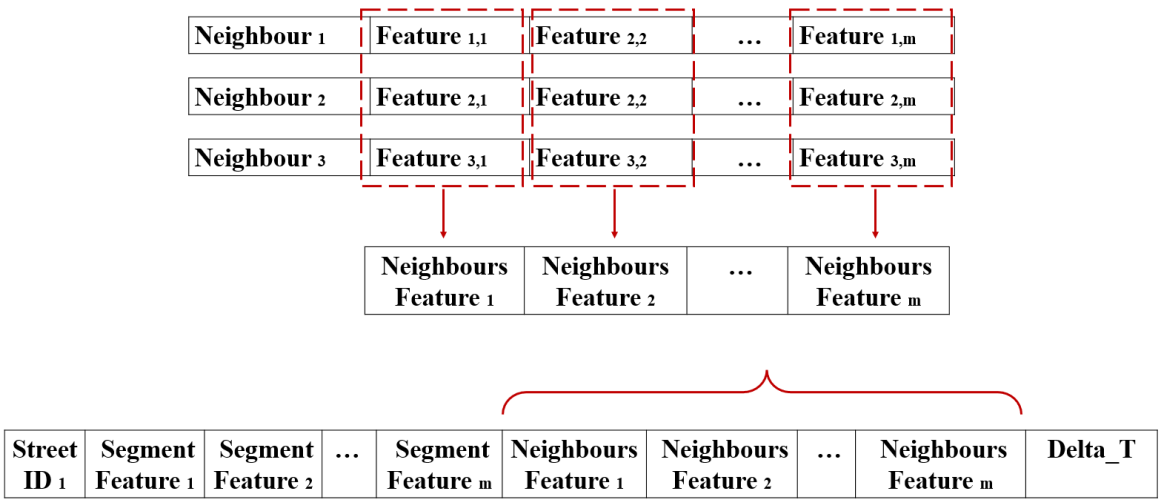


Figure 9: Separate Means Feature Aggregation

3.5 Model Training

After aggregating the features for each road segment, the resulting dataset is used to train a machine-learning model. As explained erlier, in this study, two main model types are used: Random Forest and Feedforward Neural Networks. Both models depend heavily on their hyperparameters which are configuration settings that define how the model learns. For example, for the case of a tree-based model (RF) an important parameter would be the number of decision trees or their depth, with the full list of the parameters being as follows:

- Number of trees
- Maximum number of features
- Maximum tree depth
- Minimmum number of samples per split
- Bootstrap

For the case of a Neural Network, the training hyperparameters are signifucantly different due to a distinct neture of the algorithm. Instead of the characteristics of the trees, these hyperparameters define how neurons are triggered and prcessed:

- Activation function
- Batch size
- Number of epochs
- Number of layers
- Number of neurons per layer
- Optimizer
- Kernel initialization

Choosing good hyperparameters can significantly improve the predictive performance of a model, while poor choices can lead to underfitting, overfitting, or unnecessarily long training times. To automatically identify the most suitable hyperparameter settings, a Genetic Algorithm (GA) is used. GA is an optimisation technique inspired by natural selection: it starts with a random population of candidate solutions (i.e., different hyperparameter combinations), evaluates their performance, and iteratively evolves them through selection, crossover, and mutation to find better solutions over time.

Initial tests revealed that the optimal model configuration depends on the spatial scope used for defining neighbourhoods (either via geographic radius or graph-based distance). Therefore, the GA was executed separately for different radii and node thresholds to analyse how neighbourhood size and shape affect predictive performance and the hyperparameters.

The optimisation process aims to maximise the model's R^2 score, which is a standard measure of how well the predicted values match the true temperature differences. Therefore, the objective function is defined as follows:

$$\max_{\theta} R^2(y, f(X;\theta)) \tag{33}$$

Where:

- y is the target variable, representing the temperature difference ΔT,
- X is the input feature matrix, created using one of the four aggregation strategies,
- $f(X;\theta)$ denotes the machine-learning model (RF or NN) parameterised by hyperparameters θ,
- R^2 is the coefficient of determination, which serves as the fitness score for GA.

This setup ensures that the model is not only optimised for performance but also adapted to the specific spatial context defined by the neighbourhood structure.

Training Procedure :

For each GA iteration:

1. A new set of hyperparameters was selected and applied to the chosen model.

2. The dataset was split into training (80%) and testing (20%).
3. The model was trained, and its R^2 score on the test set was computed.
4. The best-performing individuals were selected for crossover and mutation.

The final optimised hyperparameters were selected based on the individual achieving the highest R^2 score. This GA-based optimization allowed for an efficient search over a high-dimensional hyperparameter space while systematically exploring the effect of spatial scale on model performance.

Model Development Workflow :

To develop a working mode, the elements of the proposed method described above should be integrated into a coherent workflow. Figure 10 summarises all the order and steps that are needed to develop and test a potentially enhanced model for UHI assessment.

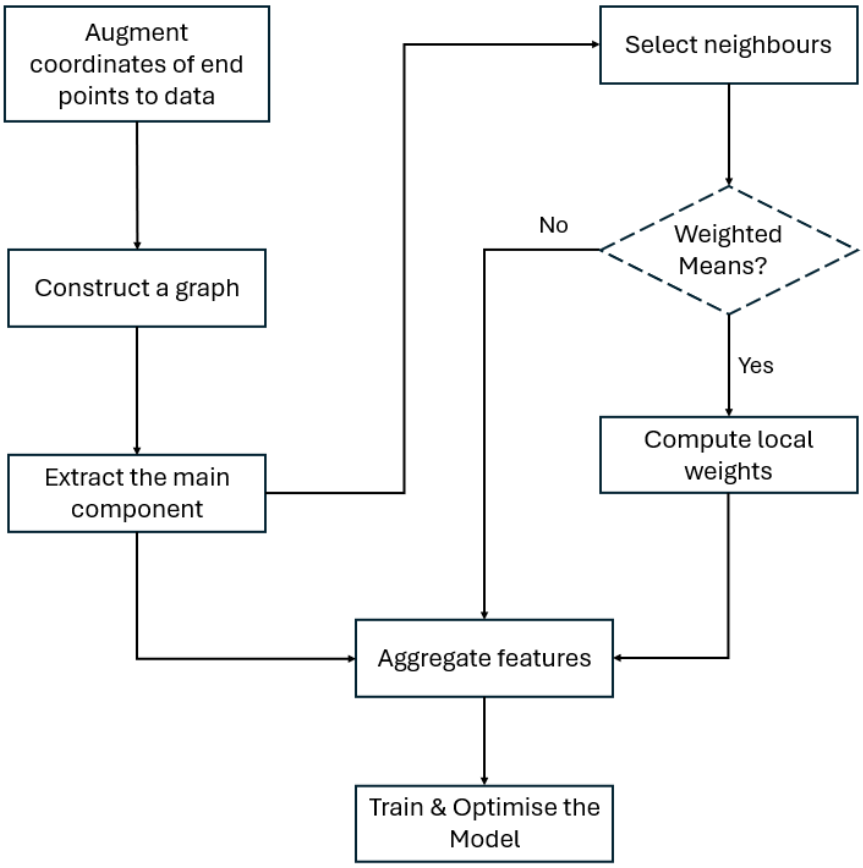


Figure 10: Workflow of a proposed method

The first steps in the process are constructing the graph and extracting its main component, which is altogether possible due to initial augmentation of the coordinates of the street segments' endpoint to data. Since the constructed graph stores both the network (edges, nodes) and feature (as edge attributes) data, except for the aggregation, it is also used to select the neighbours based on a distance or network threshold. In other words, once constructed, the graph is used as the sole source data for all the feature extraction as well as neighbour selection processes. Depending on whether the aggregation method is the Weighted Means or not, the local weights should be computed based on the neighbour selection. Finally, the selection of neighbour (with the weights) together with the feature extraction from the graph for the chosen neighbours enable feature aggregation followed by the model training and optimisation.

4 Results

4.1 Genetic Algorithm

Each radius and node threshold was tested separately, with a full GA optimization cycle applied to each. Both lists were constructed with a higher density for the lower and higher range of values and a lower density for the mid-range. This is done to balance the insightfulness of the analysis and processing efficiency since each GA iteration is computationally demanding.

The GA was executed for multiple radii r to explore the significance of spatial scale, with:

$$r \in \{100, 200, 300, 400, 500, 600, 700, 1000, 2000, 3000, 4000, 5000, 6000, 7000, 8000, 8500, 8700\} \text{ meters}$$

Moreover, similar GA was run for multiple nodes g to:

$$g \in \{0, 1, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 36, 52, 68, 84, 100, 125, 150, 175, 200, 205, 210, 215\} \text{ nodes}$$

Evolutionary Strategy :

The GA parameters used differed for RF and NN due to computational efficiency considerations, Namely, the population size in Neural Network was set to 20 individuals instead of 100, as can be seen in Table 5.

Table 5: GA Hyperparameter Settings for RF and NN

Parameter	**Random Forest (RF)**	**Neural Network (NN)**
Population size	100	20
Generations	50	50
Crossover Probability ($CXPB$)	0.8	0.8
Mutation Probability ($MUTPB$)	0.5	0.5
Stopping threshold (No improvement for R^2)	5 generations	5 generations

Training Hyperparameters :

For each GA run, the training hyperparameters were defined. For the Random Forest, the key hyperparameters were borrowed from the research of Pena [3], as can be seen in Table 6.

Table 6: Hyperparameter Ranges for Random Forest

Hyperparameter	**Search Range**
Number of trees (n_{trees})	{10, 30, 50, 100, 150, 200}
Max features (f_{max})	{sqrt, log2}
Max depth (d_{max})	{10, 20, 30, 50, 100}
Min samples per split (s_{min})	{2, 4, 6}
Bootstrap (b_{opt})	{True, False}

Similarly, for the Neural Network training, a separate set of hyperparameters was defined, as can be seen in Table 7.

Table 7: Hyperparameter Ranges for Neural Networks

Hyperparameter	**Search Range**
Activation function	{ReLU, ELU, Tanh, Sigmoid}
Batch size	{10, 50, 100}
Number of epochs	{50, 100, 150, 200, 250}
Number of layers	{1, 2, 3, 4}
Number of neurons per layer	{50, 100, 200, 300, 400, 500}
Optimizer	{Adam, Adagrad, RMSprop, SGD}
Kernel initialization	{Uniform, Normal}

Aggregation Hyperparameters :

The Weighted Means aggregation method has additional parameters related to the weighted aggregation function (see Table 8) that also needed to be optimised.

Table 8: Aggregation Hyperparameters for Experiment 4

Hyperparameter	**Search Range**
α (Weighting factor for shortest path)	{0, 0.25, 0.5, 0.75, 1}
β (Complementary weight)	$\beta = 1 - \alpha$

4.2 Random forest

4.2.1 Predictive Performance

The first set of simulations was run for the radius-based neighbour selection. The 4 feature aggregation scenarios in combination with 17 different radii yielded 68 R^2 values, as can be seen in Figure 11. Within the simulation, each GA cycle took around 1 hour and contained 6 to 7 generations with the exception of 10 generations for $R = 600m$ based on an early non-improvement threshold. Remarkably, Weighted Means converged less quickly and had 6 to 13 generations in a cycle. This resulted in around 45.5 hours of continuous computation time to obtain the graph below. The insights on GA convergence and computational efficiency are presented in the current paper for completeness of the narration and to ensure transparancy in mantaining the main advantage of the data-driven models over physical simulations.

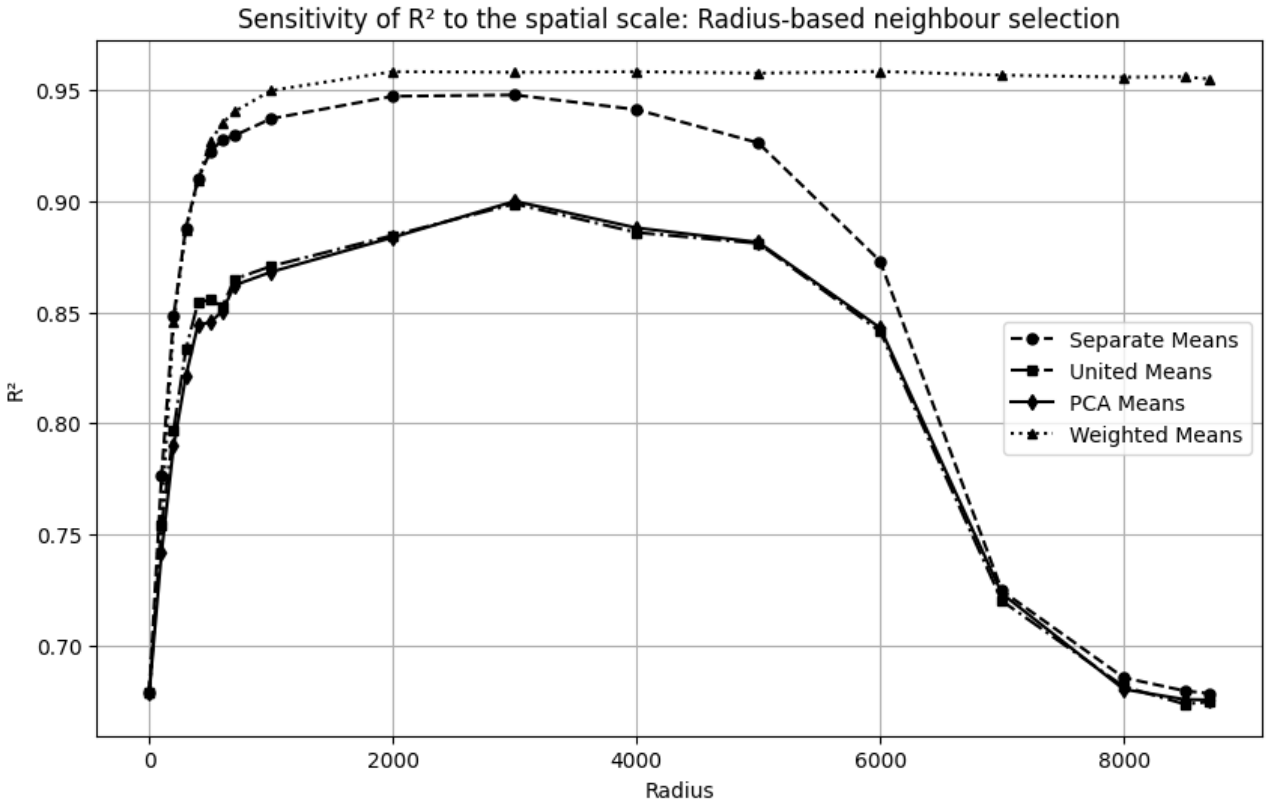


Figure 11: Sensitivity of R^2 to the spatial scale: Radius-based neighbour selection

The Radius-Based method generally achieved slightly higher peak performance compared to its enighbour-based counterparts and maintained predictive power over a much larger distance. After reaching the optimal radius, there was an extended plateau where predictive performance remained stable. In contrast, the Node-Based method showed a sharper decline in performance after its peak, indicating it is more sensitive to spatial scale.

For each feature aggregation, the hyperparameters corresponding to the GA optimisation results were found per radius. The separate results for the Separate Means, United Means, PCA Means and Weighted Means,

neighbour selection method can be seen in Tables 12, 13, 14 and 15 respectively in Appendix B.1. The Table 9 summarises the performance per radius among all four feature aggregation methods.

Table 9: R^2 for Aggregation Methods under Radius-Based Neighbour Selection (Apeldoorn)

Radius [m]	**Separate means**	**United means**	**PCA means**	**Weighted means**
0 (Baseline)	0.6786	0.6786	0.6786	0.6786
100	0.7762	0.7537	0.7415	0.7568
200	0.8483	0.7971	0.7898	0.8460
300	0.8881	0.8333	0.8213	0.8869
400	0.9100	0.8544	0.8445	0.9095
500	0.9224	0.8558	0.8454	0.9271
600	0.9278	0.8526	0.8506	0.9352
700	0.9297	0.8650	0.8622	0.9405
1000	0.9373	0.8708	0.8682	0.9500
2000	0.9474	0.8847	0.8838	0.9585
3000	0.9480	0.8989	0.9001	0.9582
4000	0.9414	0.8860	0.8881	0.9585
5000	0.9264	0.8811	0.8816	0.9578
6000	0.8731	0.8414	0.8430	0.9586
7000	0.7245	0.7199	0.7232	0.9569
8000	0.6853	0.6813	0.6801	0.9560
8500	0.6794	0.6734	0.6755	0.9562
8700	0.6782	0.6744	0.6753	0.9552

United Means and PCA Means were tested under the hypothesis that dimensionality reduction would improve performance by eliminating redundant information and reducing noise. However, the results showed the opposite effect: both methods consistently underperformed compared to Separate Means and Weighted Means. United Means averaged the original segment's features with the mean of its neighbours, effectively smoothing out spatial variation, which led to a loss of fine-grained spatial detail. PCA Means, which applied Principal Component Analysis (PCA) to reduce feature space dimensionality, similarly decreased the information value of the training set by compressing features into fewer dimensions. The results suggest that reducing dimensionality in this context sacrifices essential local and global heterogeneity, leading to weaker predictive performance.

In contrast, Separate Means and Weighted Means performed similarly and were the most effective feature aggregation strategies, with the latter showing slightly higher R^2 values. Both methods retained twice as many features as United Means and PCA Means, preserving more information about local and neighbouring conditions. Separate Means treated neighbours' features independently by averaging each feature type separately, while Weighted Means assigned varying importance to neighbours based on proximity and network structure.

Similarly, for the node-based neighbour selection, the 4 feature aggregation scenarios in combination with 24 different radii yielded 96 R^2 values, as can be seen in Figure 12. Analogous to the radius-based neighbour selection, the Separate Means, United Means and PCA Means show a similar computational efficiency with 6-7 generations needed for convergence, while the Weighted Means method alone accounts for over 50% of the running time, resulting in a grand total of around 60 hours.

Analogously, the feature aggregation method and the hyperparameters corresponding to the GA optimisation results were found per number of nodes as a search threshold. The separate results for the Separate Means, United Means, PCA Means and Weighted Means, neighbour selection method can be seen in Tables 16, 17, 18 and 19 also in Appendix B.1. The Table 10 summarises the performance per radius among all four feature aggregation methods.

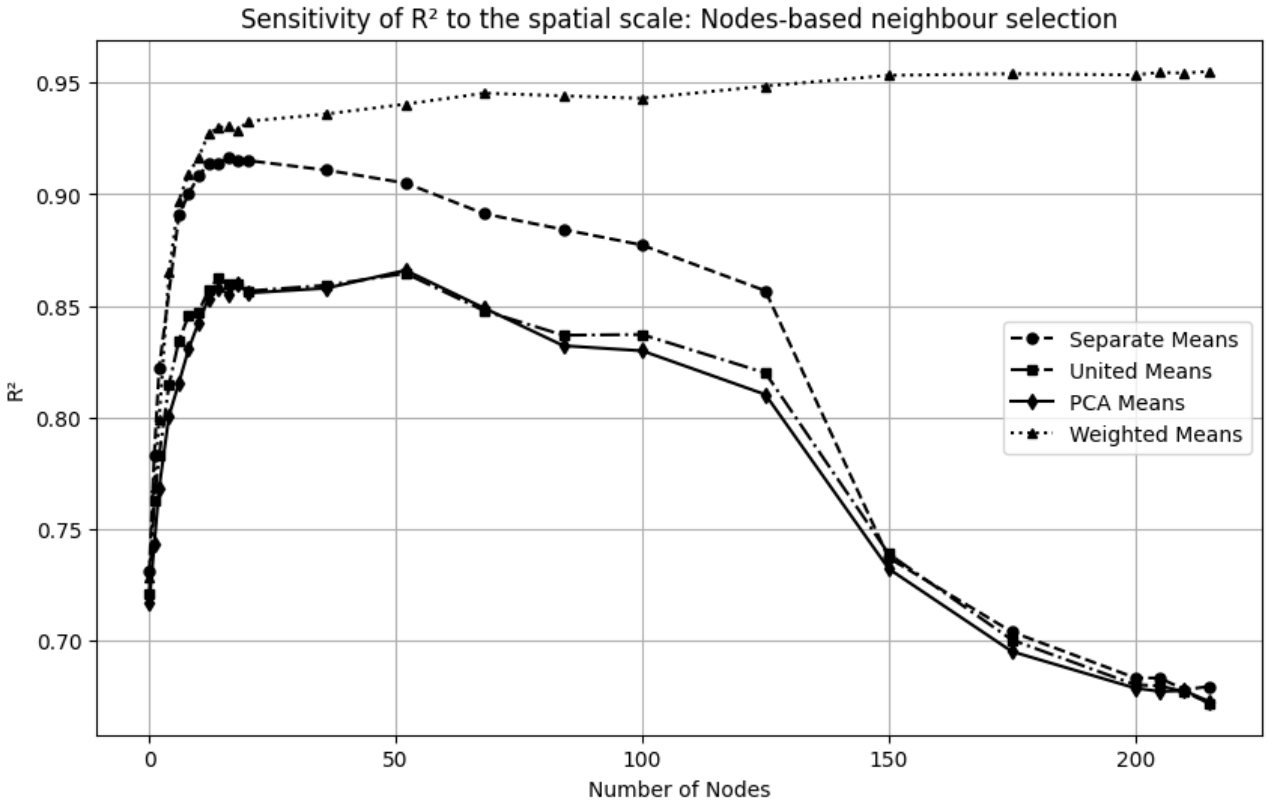


Figure 12: Sensitivity of R^2 to the spatial scale: Node-based neighbour selection

Table 10: R^2 for Aggregation Methods under Node-Based Neighbour Selection (Apeldoorn)

Number of Nodes	**Separate means**	**United means**	**PCA means**	**Weighted means**
Baseline	0.6786	0.6786	0.6786	0.6786
0	0.7307	0.7211	0.7171	0.7280
1	0.7826	0.7626	0.7428	0.7689
2	0.8221	0.7829	0.7684	0.7991
4	0.8692	0.8146	0.8007	0.8649
6	0.8905	0.8344	0.8154	0.8972
8	0.9003	0.8457	0.8308	0.9093
10	0.9082	0.8473	0.8425	0.9166
12	0.9136	0.8569	0.8533	0.9271
14	0.9135	0.8626	0.8577	0.9297
16	0.9163	0.8597	0.8553	0.9307
18	0.9152	0.8595	0.8599	0.9287
20	0.9151	0.8566	0.8557	0.9327
36	0.9108	0.8592	0.8579	0.9360
52	0.9050	0.8645	0.8659	0.9404
68	0.8912	0.8477	0.8489	0.9453
84	0.8842	0.8368	0.8321	0.9441
100	0.8772	0.8371	0.8298	0.9430
125	0.8567	0.8199	0.8102	0.9485
150	0.7370	0.7388	0.7321	0.9533
175	0.7038	0.7002	0.6950	0.9540
200	0.6832	0.6801	0.6786	0.9535
205	0.6831	0.6796	0.6771	0.9546
210	0.6779	0.6770	0.6774	0.9543
215	0.6793	0.6718	0.6726	0.9551

The difference between Separate Means and Weighted Means becomes especially pronounced in the Node-Based approach. It can be seen that in contrast to the radius-based selection, during the node-based alternative, Separate Means started experienceing a drop in performance at relatively small values of nodes. It can be expect that this happens similarly because averaging over an increasingly broad spatial range diluted meaningful local variations.

However, the unexpected loss of robustness of this method to the mid-range spatial scale might suggest that the network effects are strongest at a block level in the Apeldoorn dataset. The Weighted Means did not experience the same decline due to its ability to preserve local spatial gradients by assigning higher importance to closer neighbours. As a result, Weighted Means continued to perform well across a wider range of the neighbour selection threshold (in this case, nodes).

Note that for the radius-based neighbour selection methods, the simulation at radius $R = 0m$ represent the baseline model similar to the result of Pena Acost [3] used to evaluate the performance of the proposed method. Namely, the Random Forest model inputs only the features of the street segment itself, i.e. without any definition of neighbours or aggregation of features. However, for the node-based neighbour salection, the $g = 0nodes$ corresponds to the case where street segments with the direct connection to a given segement are included. For both cases, the baseline for comparison is the same since this case does not involve any neighbour selection or feature aggregation steps.

4.2.2 Feature Importance

Another outcome of the Random Forest training is a feature importance distribution. Since the current paper focuses on modelling UHI phenomenon at different spatial scales, it was also studied how the feature importance fluctuates depending on the feature aggregation scale. For this purpose, the model has been run for multiple radii using the Radius-based neighbour selection complemented by the best performing Separate Means and Weighted Means feature aggregation methods.

For visualisation purposes, 5 to 6 features were selected that form the general pattern observed on a complete graph initially. Moreover, the feature importance pertaining to the segment and to its neighbours are deliberately split into two different plots for further analysis. Following the procedure described above, the feature importance fluctuation for the segment's feature using the Separate Mean feature aggregation method can be seen in Figure 13, while the fluctuations across different radii for the neighbours can be observed in Figure 14

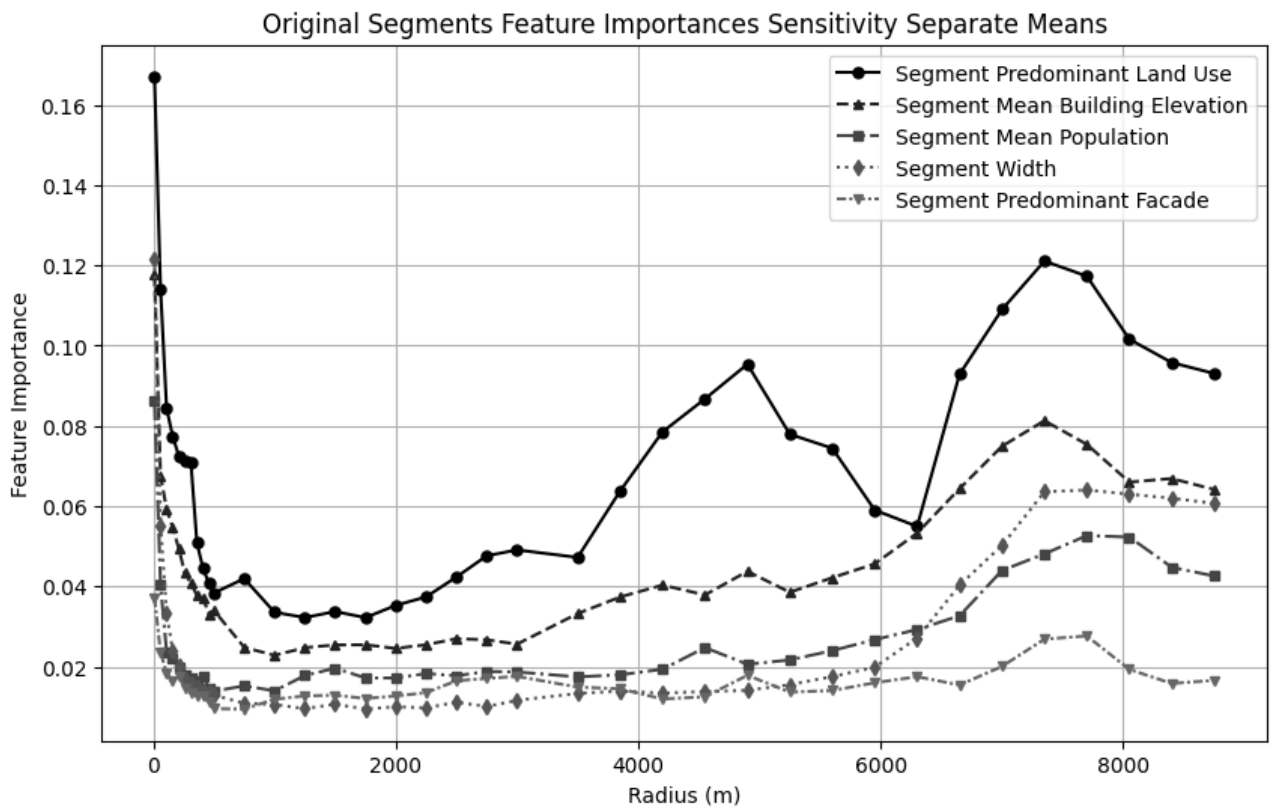


Figure 13: Original Segments Feature Importances Sensitivity Separate Meansl

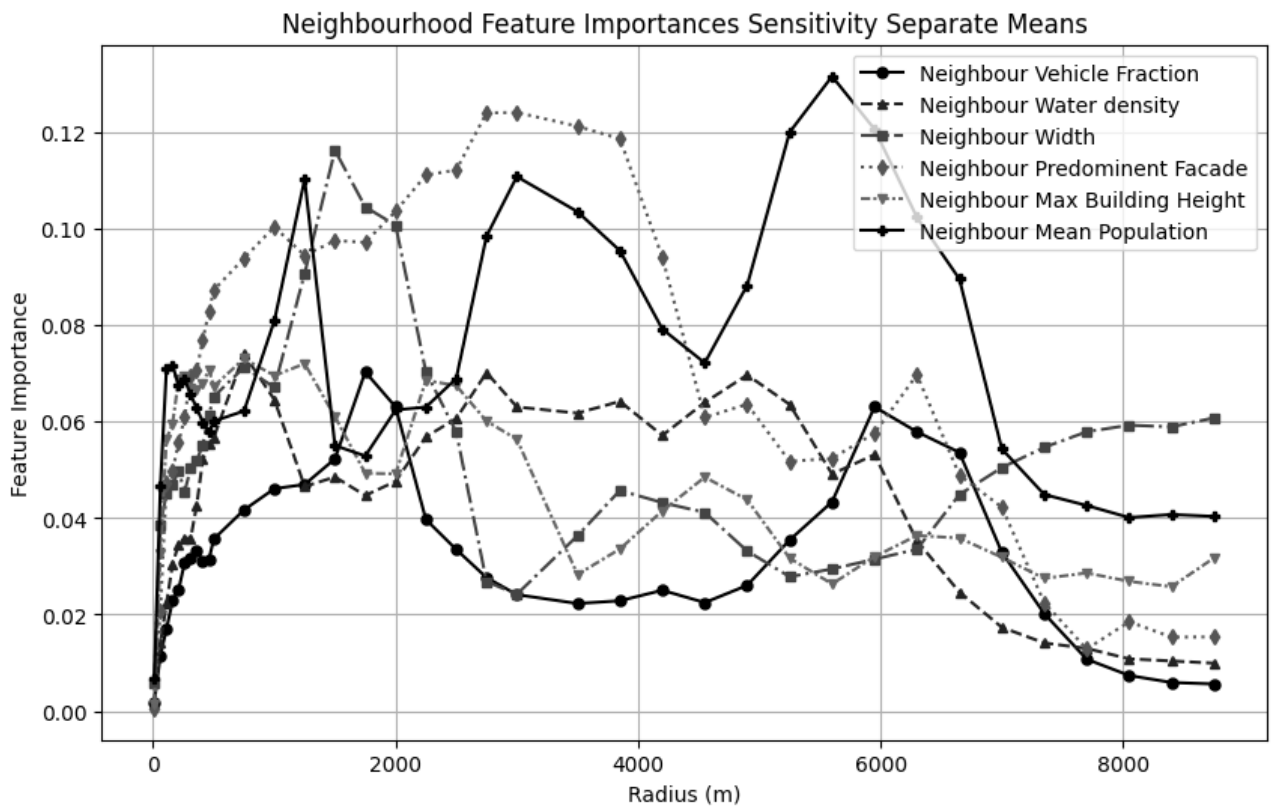


Figure 14: Neighbourhood Feature Importances Sensitivity Separate Means

Similarly, for the Weighted Means feature aggregation, the result for the features of the segments is shown in Figure 15.

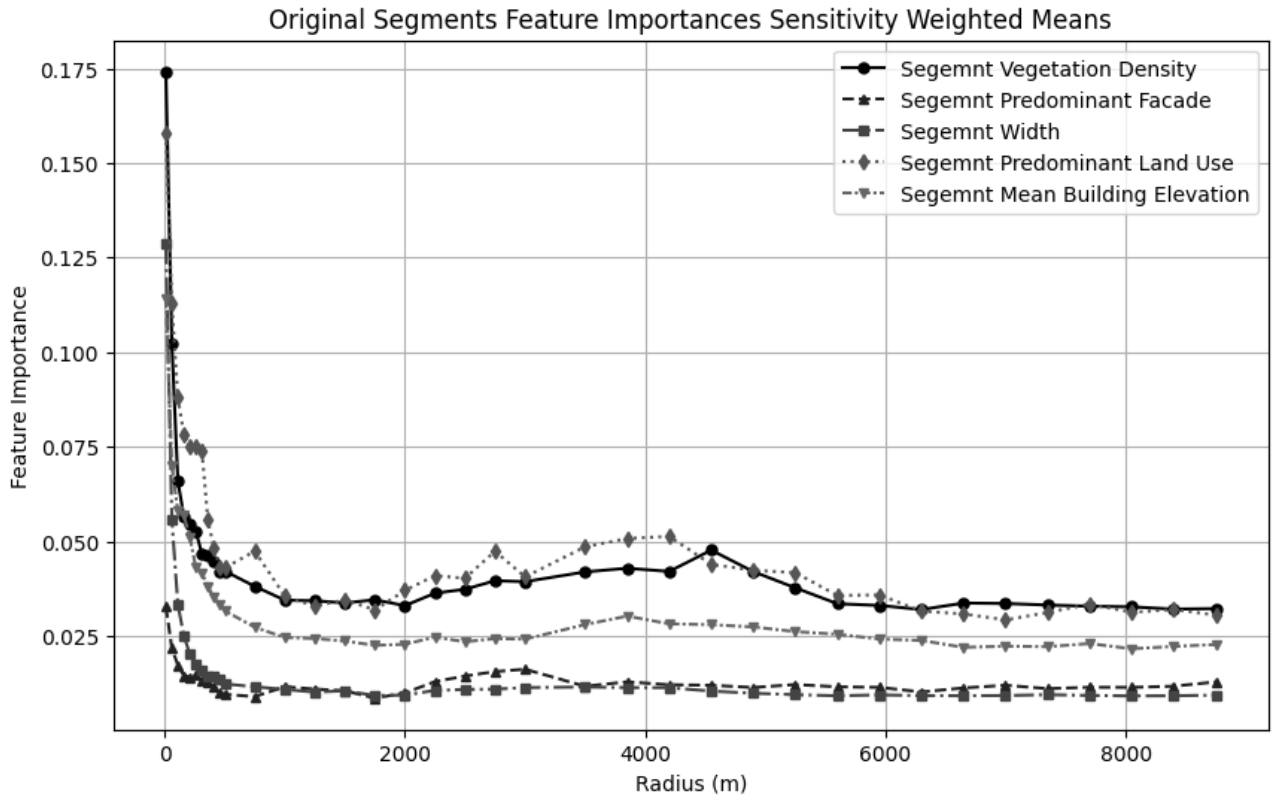


Figure 15: Original Segments Feature Importances Sensitivity Weighted Means

Lastly, The feature fluctuations across different radii for the neighbours using the Weighted Means can be observed in Figure 16

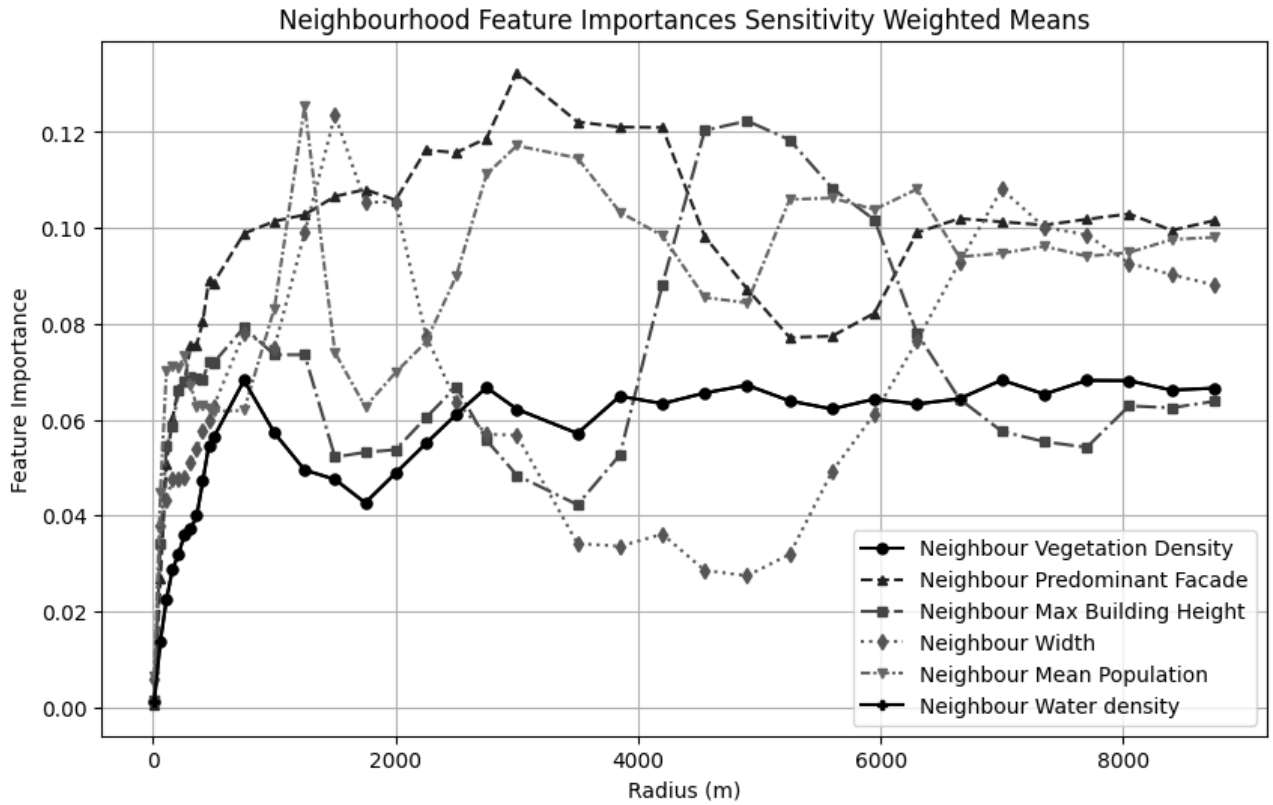


Figure 16: Neighbourhood Feature Importances Sensitivity Weighted Means

For the original segment features, vegetation density and predominant land use dominate the predictions at small spatial scales (low radii). This indicates that on a localised street or block level, immediate land cover characteristics and green coverage are key determinants of UHI variations. However, as the spatial scale increases, the importance of these features gradually fades along with other segment-based predictors. This pattern suggests that beyond a certain distance, the direct influence of the segment itself becomes less relevant as broader spatial patterns take over.

Interestingly, in the Separate Means approach, segment features regain importance at very large radii. This phenomenon is likely a result of over-averaging neighbouring information, making the predictions increasingly homogeneous. When the radius becomes large enough, the averaged neighbour contributions become nearly identical across the dataset, rendering them uninformative. Consequently, the model begins relying on segment-level features again, even if their relevance should theoretically diminish at larger scales. This behaviour is not observed in Weighted Means, which further supports the proposition above and the hypothesis that a gradual weighting of neighbouring information prevents the effect of over-averaging.

The fluctuating patterns observed in neighbourhood-level feature importance should not be interpreted as evidence that each feature has a distinct "optimal" spatial scale at which it exerts maximal influence. Rather, these fluctuations are driven by the changing distributional structure of each feature as spatial averaging is applied. Some features become spatially homogeneous more rapidly than others. In other words, their values become increasingly similar across all segments as the radius grows. This homogenisation reduces the feature's variance, thereby limiting its capacity to explain variations in UHI, and consequently lowering its importance score in the model. Other features may retain more spatial heterogeneity across larger distances and thus remain more informative over a wider range of radii.

This explains why certain features peak in importance at specific radii, which happens not because they are inherently more relevant at those scales, but because that is the point at which they still retain enough variation to explain differences in UHI outcomes. The peaks and subsequent declines in importance are thus the evidence of how spatial aggregation transforms the input features, not necessarily of real-world scale-dependent behaviour. Once the feature becomes too uniform, its predictive power naturally diminishes.

4.3 Neural Network

In principle, the same set of 8 simulations needs to be conducted for the Neural Network training algorithm. However, the preliminary exploration iteration revealed two facts: the NN-based training is significantly less computationally efficient and its prediction performance is consistently lower compared to a Random Forest. Namely, the preliminary testing was performed on the Separate Means feature aggregation method since it was noticed from the RF-based model results obtained earlier that this method is the second best in terms of prediction performance while being significantly more computationally efficient than the Weighted Average Mean aggregation that shows to have the highest prediction capability. The neighbour selection method was chosen to be radius-based because of its higher prediction capacity and overall more stable performance. Due to encountered computational capacity challenge, a selection of 7 iterations was run on the set of radii representing the range of the city with the lower granularity than initially proposed:

$$r \in \{100, 500, 1000, 3000, 4000, 5000, 8000\} \text{ meters} \tag{34}$$

The results of the preliminary iterations can be seen in Table 11. Indeed, it was noticed that the Neural Network-based training requires more optimisation iterations to converge, ranging from 6 to 18 cycles. On top of the nature of NN training that requires more time to train, this exacerbates the computational capacity challenge, resulting in 4 to 12 hours of runtime for a single radius. Therefore, scaling the simulation of the NN-based training to 68 and 96 threshold values in the case of the radius- and node-based neighbour aggregation would result in over 476 and 672 minutes needed respectively to obtain the graphs similar to those presented for the RF-based training.

Moreover, this approximation is based on the data from the Separate Means feature aggregation method, which excludes significant decrease in computational efficiency during Weighted Mean simulation. Hence, based on the analogy of the RF-based model, it can be expected that the actual required run time for the NN-based model would be greater by a factor that lies in the neighbourhood of 2.

Table 11: Best Hyperparameters per Radius for Neural Network, Separate Means

Radius [m]	Best R^2	Activation	Batch Size	Dropout Rate	Epochs	Kernel Init	Layers	Neurons	Optimizer
100	0.7290	elu	50	0.0	150	uniform	4	341	adam
500	0.8796	tanh	100	0.1	250	normal	3	475	adam
1000	0.8855	tanh	100	0.1	250	uniform	3	438	adam
3000	0.8475	tanh	100	0.1	200	uniform	4	416	adam
4000	0.7765	relu	10	0.0	200	normal	4	298	rmsprop
5000	0.7114	elu	100	0.0	200	uniform	4	107	adam
8000	0.5957	tanh	50	0.1	250	normal	3	340	adam

Furthermore, except for the computational efficiency challenge, the NN-based models show a consistently lower prediction performance compared to the same neighbour selection and aggregation method but trained on a RF algorithm, as can be seen in Figure 17. Therefore, it was decided not to conduct the entire set of proposed simulations on the NN-based model as they showed suboptimal performance due to preliminary testing.

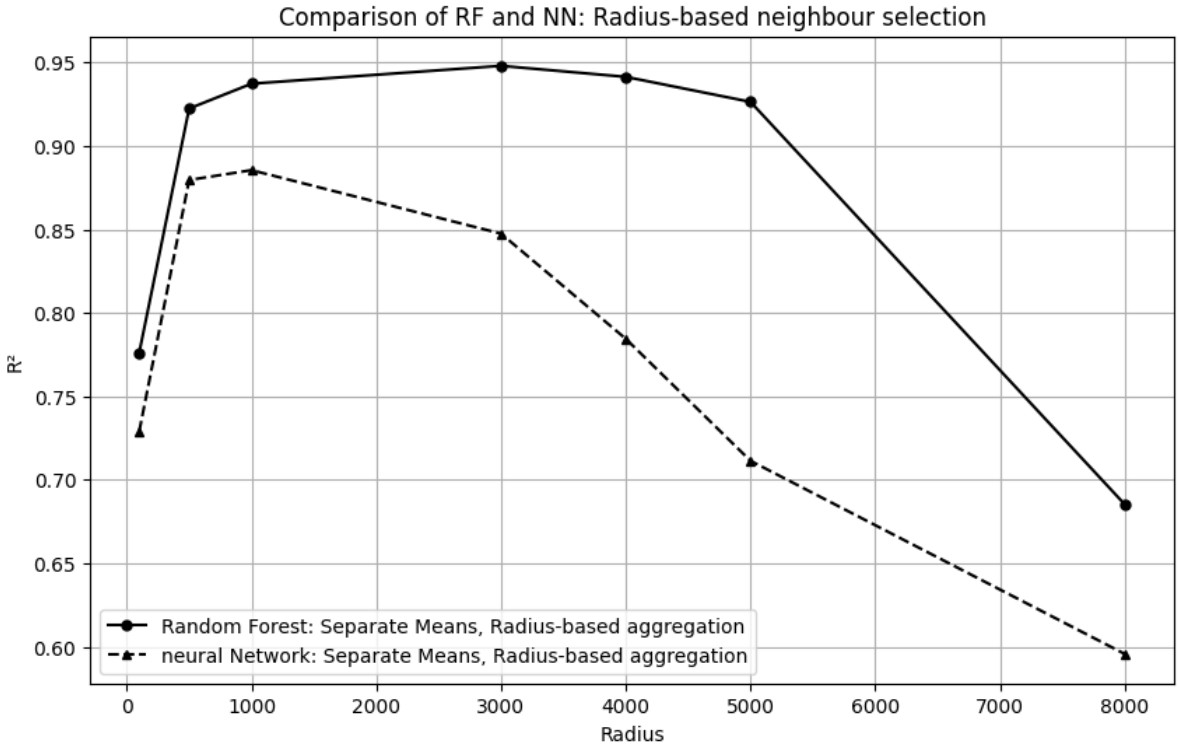


Figure 17: Comparison of RF and NN: Radius-based neighbour selection

4.4 Linear Regression

Additionally, the linear models have been tested according to the proposed adjustment of the spatial scale. Figure 18 shows the result of the SLX simulation for various radii. The peak value of $R^2 = 0.58$ is attained in the proximity of $R = 1600m$, whereas the baseline model corresponding to a linear regression without considering neighbours is simulated at $R = 0m$ and yields $R^2 = 0.39$. Note that, unlike the Random Forest of Neural Network, the Linear Regression does not require hyperparameter tuning (GA algorithm).

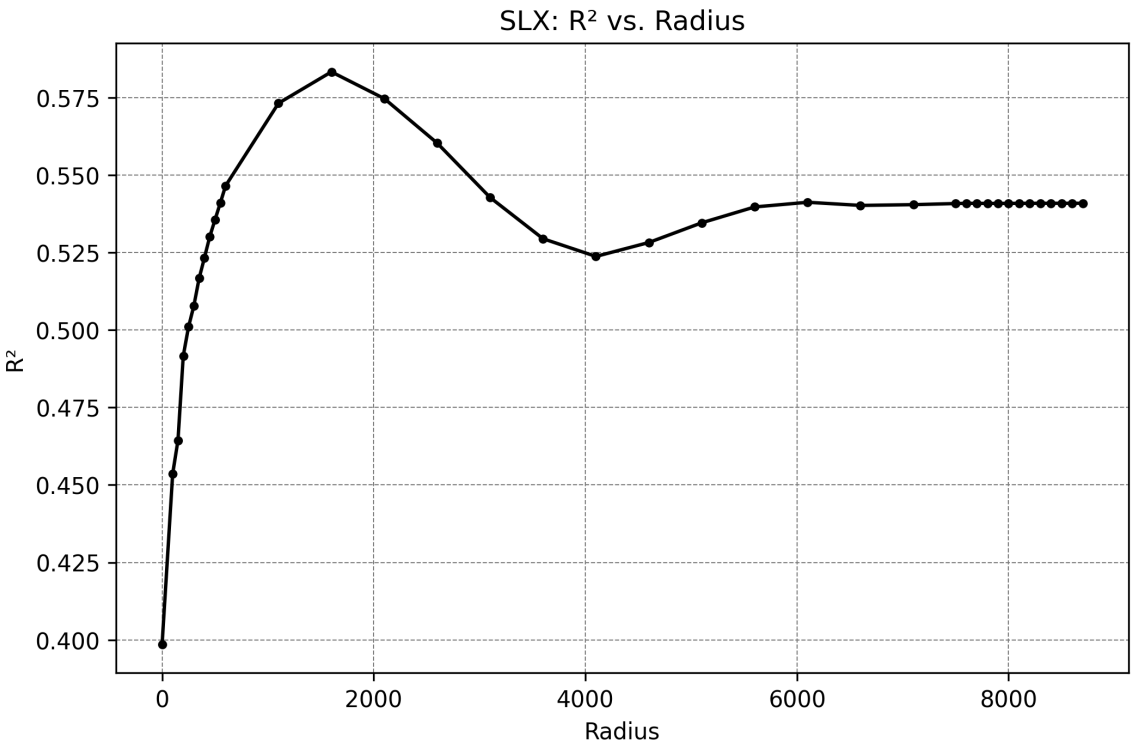


Figure 18: SLX: R^2 vs Radius

As expected, linear regression models did not match the performance of ensemble or deep learning approaches, reinforcing the inherently nonlinear nature of UHI relationships. However, the significant increase in R^2 from 0.39 in the baseline model to 0.58 in the Spatial Lag Model (SLX) suggests that incorporating spatial dependencies in linear models does provide meaningful improvements, even if they remain suboptimal compared to more advanced machine learning techniques.

4.5 Testing the Pipeline

The developed pipeline was tested on 3 datasets presented in Section 1.2.5. The present chapter elaborates on the findings for each dataset.

4.5.1 Rotterdam

In the course of data processing, it emerged taht both Rotterdam and Montreal datasets are highly fragmented into many disconnected patches of street segments. This makes extraction of the main component of the graph meaningless since this eliminates most of the data. Hence, this step of the pipeline cannot be executed on the given datasets. However, this also implies that the node-based neighbour selection cannot be executed either. Therefore, the most applicable method is a radius-based neighbour aggregation followed by the two most competitive feature aggregation methods: Separate Means and Weighted Means. The same conclusion applied to the bike-collected data, since it includes only a small selection of street segments forming a closed curve.

Since every city has a different size, the spatial scale variation was adjusted. For Rotterdam, the set of radii was set to:

$$r = \{100, 200, 300, 400, 500, 750, 1000, 1250, 1500, 1750, 2000, 2500, 3000, \\ 3500, 4000, 5000, 7000, 9000, 11000, 13000, 15000, 20000, 30000, 43000\}$$

For Rotterdam, the above simulation with 2 feature aggregation methods yields the following results (see Figure 19).

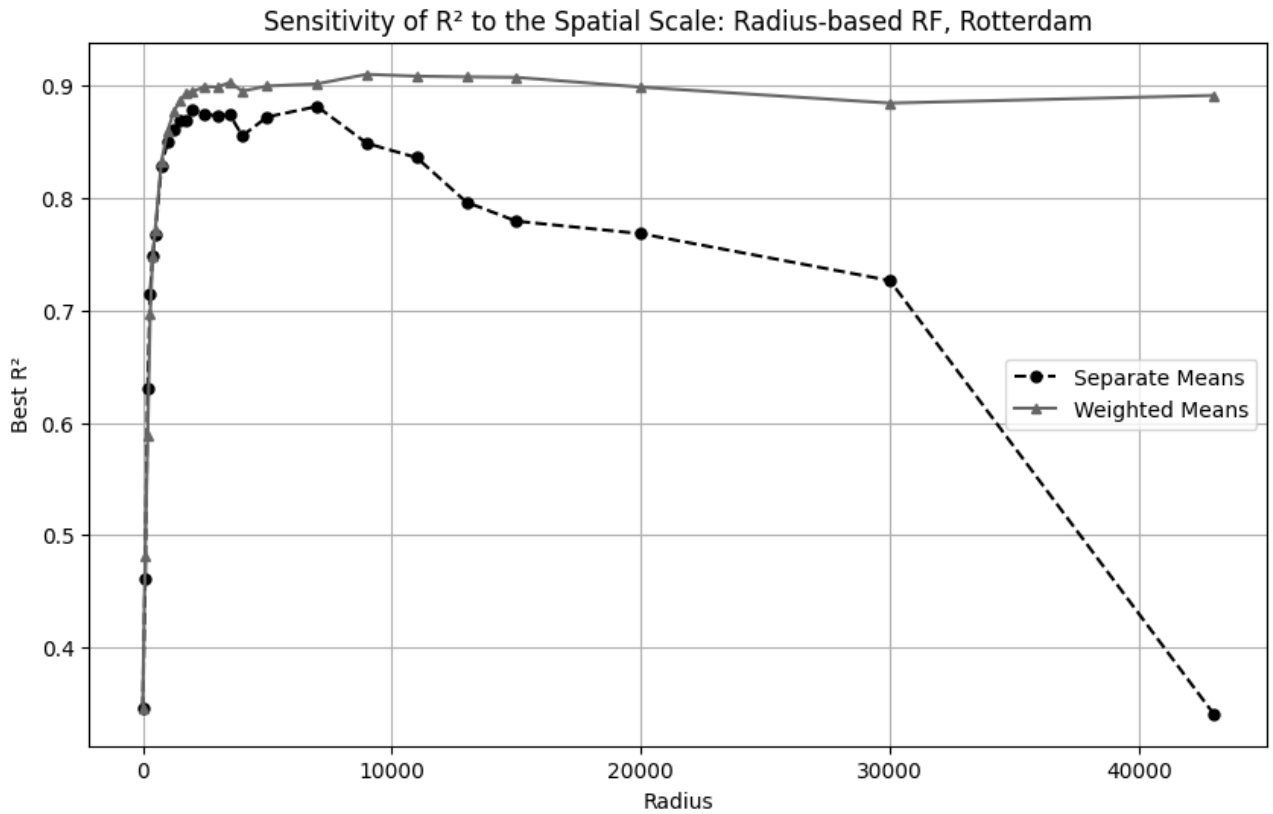


Figure 19: Sensitivity of R^2 to the Spatial Scale: Radius-based RF, Rotterdam

Following the pipeline, the hyperparameters corresponding to the GA optimisation results were found per radius. As can be seen in Appendix B.2, the results for the Separate Means neighbour selection method are presented in

Table 20 while for the Weighted Means, the best hyperparameters per radius can be seen in Table 21.

For Rotterdam, the peak value of $R^2 = 0.88$ is attained in the proximity of $R = 7000m$ using Separate Means, while $R^2 = 0.91$ is achieved at $R = 9000m$ using Weighted Means feature aggregation, whereas the baseline model corresponding to a Random Forest without considering neighbours is simulated at $R = 0m$ and yields $R^2 = 0.34$. The relatively large scale needed to achieve highest perfromance suggests that in this particular city, broader spatial dependencies improve model performance. A potential explanation could be that Rotterdam's urban morphology changes gradually, which may require larger spatial radii for the model to effectively learn these patterns.

Moreover, it can be seen that the Separate Means feature aggregation method experiences a sharp decline after peaking at $R = 7000m$, unlike the Weighted Means approach, which maintains robust to a multiefold expansion of the spatial scale. The difference between two methods using the radius-based aggregation was not as ronounces in the earlier datasets. Supposedly, this occurs due to the range of the spatial scale of Rotterdam that is 5 times larger compared to Aperldoorn's satellite data. This further highlights that the predictive performance of the Separate Means method is only comparable with the Weighted Means at a spatial scale of less than 10.000m.

4.5.2 Montreal

Similarly, for Montreal, the radius-based neighbour aggregation followed by Separate Means and Weighted Means aggregation methods yields the following results (see Figure 20).

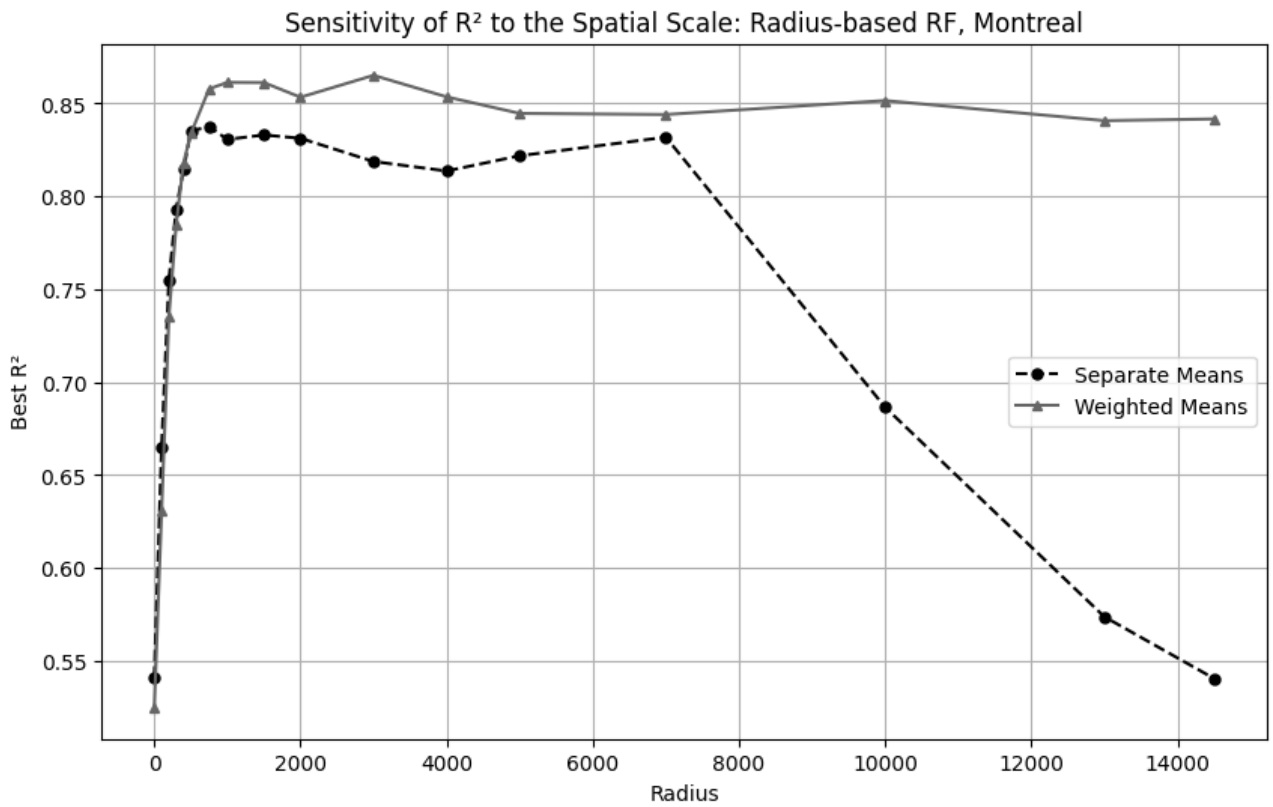


Figure 20: Sensitivity of R^2 to the Spatial Scale: Radius-based RF, Montreal

As can be seen, for Montreal, the following radii were considered as a variation of a spatial scale:

$$r \in \{0, 100, 200, 300, 400, 500, 750, 1000, 1500, 2000, 3000, 4000, 5000, 7000, 10000, 13000, 14500\} \text{ meters}$$

Moreover, the hyperparameters corresponding to the GA optimisation results per radius for the Separate Means neighbour selection method can be seen in Table 22 in Appendix B.3. For the Weighted Means, the best hyperparameters per radius can are presented in Table 23.

For Montreal, the peak value of $R^2 = 0.83$ is attained in the proximity of $R = 500m$ using Separate Means and $R^2 = 0.86$ at $R = 1000m$ using Weighted Means feature aggregation, whereas the baseline model corresponding to a Random Forest without considering neighbours is simulated at $R = 0m$ and yields $R^2 = 0.34$.

Similarly to the results of Rotterdam, the Separate Means remains a competitive feature aggregation method only at the moderate spatial scale. Interestingly, although the disance between furthermost points available in the dataset of Montreal is aproximately 4 times smaller than one of Rotterdam, the sharp decrease in performance of the Separate Means is observed at roughly the same spatial scale: $R = 7000m$. Although most likely coincidental, this finding highlights the fact that the success of application of the Separate Means method is depend on a spatial scale and the dataset properties, while the Weighted Means approach shows to be robus for both.

4.5.3 Apeldoorn-Bike data

The bike data come from an 8 km circuit forming a closed loop with 105 distinct street segments. Unlike the rest of the urban dataset, the dike data has a temporal component. Each street segment was measured multiple times across different dates and times of day. Hence, each street segment has a time series of temperature and other dynamic attributes (e.g., reference temperature, relative humidity).

Therefore, each segment can have multiple rows in the dataset, one row per measurement event. Because the target variable and dynamic features vary over time, only neighbours measured at the same Date and Part of the day can contribute. Thus, when aggregating neighbour features, not only the neighbouring segment ID but also the same time index was incorporated in processing the bike data. This resulted in 17325 distinct measurements.

Moreover, the bike data has different set of target variables, distinguishing between the Canopy and Surface temperature. Therefore, in the application of the designed pipeline, the radius-based neighbour aggregation followed by the Separate Means and Weighted Means feature aggregation methods is applied separately to the Canopy and Surface temperature targets. This yields 4 different models.

After adjusting the model to suit the nature of the bike data, the values of R^2 added hyperparameters corresponding to the GA optimisation results were found per radius. The resultant R^2 values per neighbour selection radius for the simulations where Surface Temperature was set as a target are displayed in Figure 21. Moreover, the detailed results including best hyperparameters for both used feature aggregation methods are summarised in Tables 24 and 25, as can be seen in Appendix B.4.

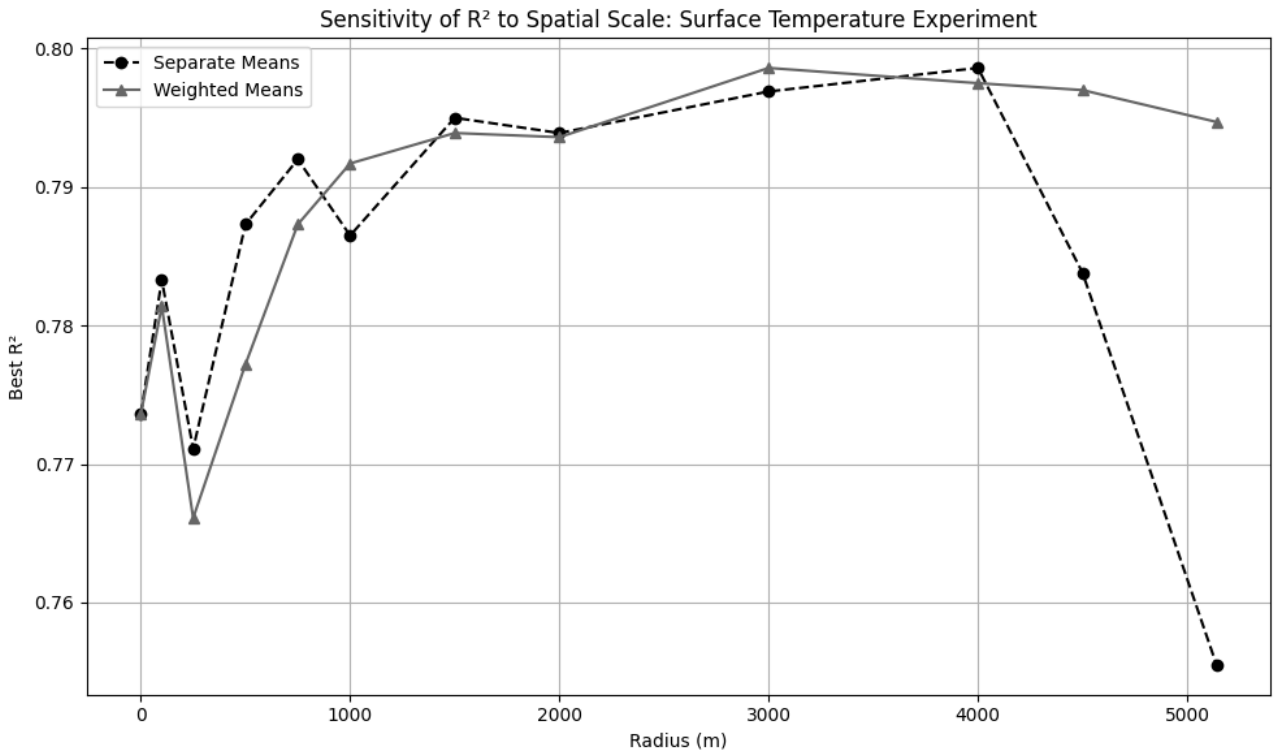


Figure 21: Sensitivity of R^2 to the Spatial Scale: Radius-based RF, Surface Temperature target

Similarly, the the simulations where Canopy Temperature was set as as a target are displayed in Figure 22. Likewise, the detailed results including best hyperparameters for both Separate Means and Weighted Means feature aggregation method are summarised in Tables 24 and 25 in in Appendix B.4.

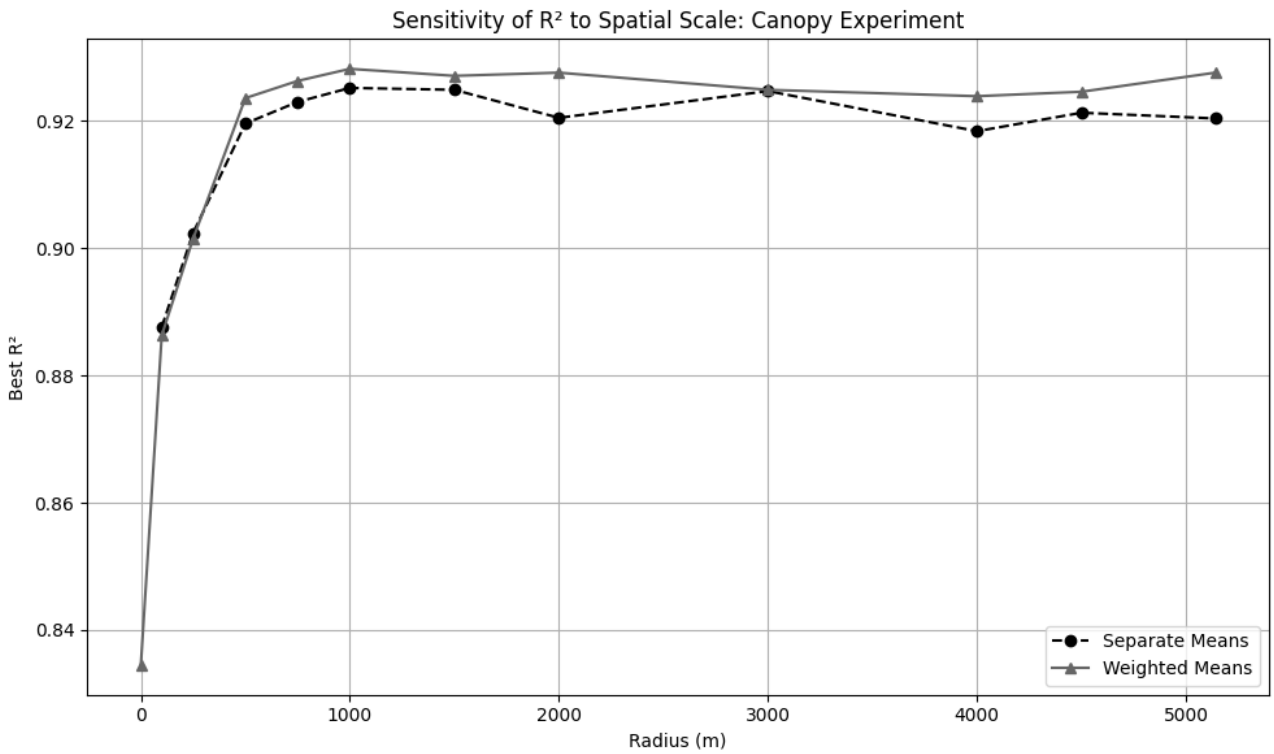


Figure 22: Sensitivity of $\mathbf{R^2}$ to the Spatial Scale: Radius-based RF, Canopy Temperature target

The peak value of $R^2 = 0.7978$ is attained in the proximity of $R = 1000m$ using Weighted Means feature aggregation with the Surface Temperature target and $R^2 = 0.9262$ at $R = 750m$ using Weighted Means with the Canopy Temperature as a target. The baseline model corresponding to a Random Forest without considering neighbours is simulated at $R = 0m$ yields $R^2 = 0.7687$ in the Surface Temperature target experiment and $R^2 = 0.8375$ in the Canopy Temperature target simulation.

It can be seen that for the surface temperature, the improvement is marginal, increasing from 0.77 to 0.8. This suggests that more neighbouring segments are needed to capture sufficient spatial variability and improve model performance. Moreover, the bike dataset required the incorporation of a temporal component into the model. This demonstrates that the pipeline can accommodate temporal data, suggesting a promising direction for future applications.

Interestingly, while land surface temperature prediction behaved as expected, with Separate Means exhibiting an initial improvement followed by a decline in performance beyond the optimal radius range (named plateau earlier), the Canopy Temperature experiment yielded much higher predictive performance and did not exhibit the same drop in accuracy at larger radii. This could indicate that canopy temperature is influenced by processes occurring at a broader spatial scale, making a larger aggregation radius more meaningful.

5 Discussion

The findings of this study examine the role of spatial dependencies in Urban Heat Island (UHI) assessment models. The results indicate that explicitly modelling spatial relationships between urban morphological and socio-economic features leads to significant improvements in predictive performance. This discussion synthesises key insights, evaluates methodological choices, and explores the broader implications of the findings.

5.1 Key Findings of Pipeline Development

5.1.1 Machine Learning Models

All the models in the scope of the current research showed a significant improvement in performance compared to their baseline counterparts that do not consider the influence of the neighbouring segments on the prediction. Hence, the study confirms that incorporating spatial interdependencies significantly enhances UHI assessment models. The modelling approached without spatial consideration performed considerably poorer, with an increase of up to 41% in performance being observed when spatial relationships were explicitly modelled (in the case of the Apeldoorn satellite data with the Radius-based neighbour selection and Weighted Means feature aggregation). Regardless of the neighbour selection, feature aggregation or a Machine Learning training method, the tested models incorporating adjacency relationships consistently outperformed baseline models. This validates the hypothesis that urban heat propagation is inherently spatially interdependent, and its prediction should not rely solely on the properties of the prediction segment itself.

One of the most critical insights is the effect of spatial scale on model accuracy. The results showed that while increasing the neighbourhood size (R) in the closest proximity sharply improved $\mathbf{R^2}$, the gains diminished significantly beyond a certain radius. For example, of e.g. 4000m-5000m for Apeldoorn. This suggests that in the case of Apeldoorn, considering the neighbours within 400m improved performance by 36% increasing $\mathbf{R^2}$ from 0.67 corresponding to the absence of neighbours to 0.92. However, at the spatial scale of, 5000m, a sharp decrease in performance was observed. This could suggest that there is an optimal spatial resolution where the signal-to-noise ratio is maximised, balancing fine-grained urban effects with broader climatic influences. Beyound these point, averaging features over more distant neighbours dilute the relevance for the considered segemnt and hence result in a less meaningful prediction. At the extreme $R = R^{\max}$, the entire city is aggregated, which foreseably results in a loss of any advantage gained by introduction of the spatial aggreagation for the prediction. Note that the numbers presented in the argument above are based on the Apeldoorn satellite data in the simulation with the Random Forest used to train the model, Separate Means feature aggregation method and Radius-based neighbour selection approach.

As explained in Section 3.2, 16 models excluding linear regression were planned to be fully deployed. However, initial testing revealed that the Neural Network consistently underperformed compared to the Random Forest. While Neural Networks are theoretically well-suited for complex pattern recognition, they require significantly more computational resources, rigorous hyperparameter tuning, and extensive data preprocessing. Despite this additional effort, their performance remained inferior to Random Forest in this study.

Conversely, Random Forest delivered high predictive accuracy with minimal tuning, demonstrating its robustness against overfitting and its ability to effectively model nonlinear relationships. The fact that Random Forest achieved superior results without requiring the extensive optimisation necessary for Neural Networks makes it a more practical choice for UHI modelling at the city scale. Furthermore, its interpretability and efficiency further justify its preference over Neural Networks, which proved to be more computationally demanding and less transparent.

5.1.2 Feature aggregation methods

Four feature aggregation methods were tested: Separate Means, Weighted Means, United Means, and PCA Means. Each approach aimed to explore how different representations of neighbouring features influence model performance. The results demonstrated clear distinctions between methods in terms of effectiveness and stability across different thresholds.

An important distinction emerged at larger radii. Separate Means suffered from a drop in performance at large R values because averaging over an increasingly broad spatial range diluted meaningful local variations. When R became too large, the model effectively averaged features across the entire city, leading to a loss of spatial resolution and a near-homogenisation of predictor values making the predictions nearly meaningless.

Weighted Means, on the other hand, did not experience the same decline in performance at larger radii. This is because Weighted Means assign higher importance to closer neighbours while still incorporating long-range influences in a controlled manner. This weighting mechanism preserved local spatial gradients, preventing the

excessive smoothing effect observed in Separate Means. As a result, Weighted Means continued to perform well across a wider range of radii and node thresholds, making it the more robust aggregation method in capturing UHI variations across different spatial scales.

The findings suggest that while Separate Means are highly effective at moderate radii, Weighted Means provide a more stable and scalable approach, particularly when broader spatial relationships need to be considered without losing localised influence.

5.1.3 Neighbour selection methods

Two neighbour selection methods were considered: Radius-Based and Node-Based selection. The results reveal key differences in their predictive power and sensitivity to spatial scale, with notable implications for model performance. The Radius-Based method generally achieved slightly higher peak performance and maintained predictive power over a much larger distance. After reaching the optimal radius, there was an extended plateau region where predictive performance remained stable, suggesting that the model continued to effectively capture spatial dependencies without significant degradation. In contrast, the Node-Based method showed a sharper decline in performance after reaching its peak. This indicates that while Node-Based selection is effective at small scales, it loses predictive stability as the number of considered nodes (intersections) increases.

The difference between Separate Means and Weighted Means is particularly pronounced in the Node-Based approach. As observed previously, Weighted Means maintain their predictive power better due to the decreasing importance of more distant neighbours, whereas Separate Means suffer from an eventual dilution effect. The sharper decrease in performance of the node-based aggregation at smaller values of the neighbour selection threshold compared to the radius-based selection exacerbated the difference between the two aggregation methods making the Weighted Means superior due to its ability to prevent the over-smoothing that occurs when too many distant nodes are averaged equally.

A critical factor influencing these results is urban morphology. Figure 23 visualises that Node-Based and Radius-Based neighbours can differ significantly in their spatial distribution. In a network-constrained environment (e.g., a city with waterways, highways, or irregular street patterns), the Node-Based approach may fail to select meaningful mid-distance neighbours due to physical barriers requiring long detours. For instance, if two segments are separated by a river or highway, they may be connected via a long bypassing route, requiring traversal through many intermediate nodes—thereby increasing the total node count and making it less likely for them to be considered neighbours. In contrast, the Radius-Based method selects neighbours purely based on Euclidean distance, ignoring network connectivity, which can sometimes provide a more meaningful spatial representation depending on the urban layout.

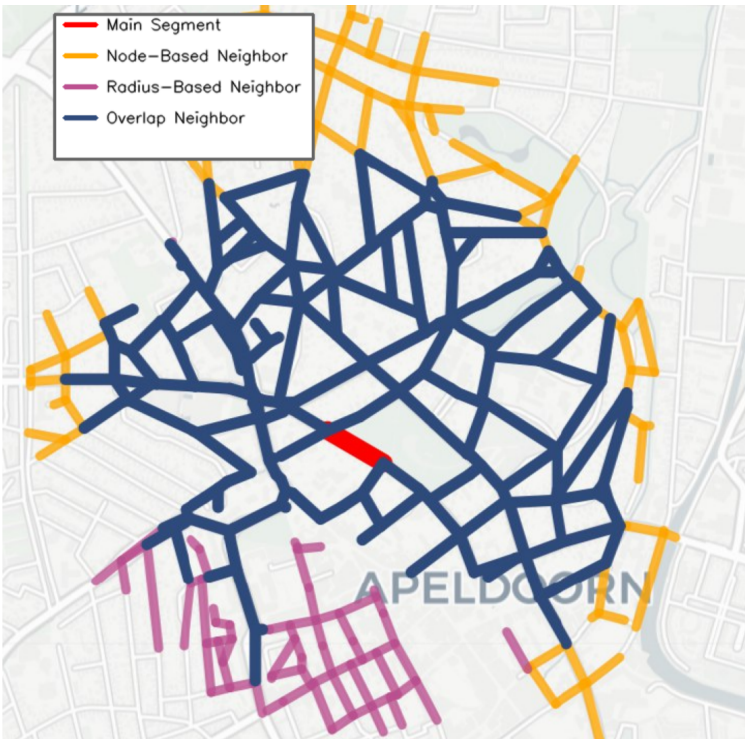


Figure 23: Comparison of Radius-based and Node-based neighbour selection

Ultimately, the choice between Radius-Based and Node-Based selection depends on the specific city's spatial structure. In highly fragmented cities with significant physical barriers, Radius-Based methods may be preferable as they ensure that heat diffusion relationships are captured even when direct road connections are absent. However, in compact urban areas where network connectivity is a stronger predictor of environmental conditions, Node-Based selection may still provide valuable insights.

5.1.4 Feature importance

Feature importance analysis reveals critical insights into how different features contribute to UHI predictions across varying spatial scales. Examining both segment-level and neighbourhood-level features using Separate Means and Weighted Means aggregation methods highlights patterns of influence, fluctuations, and stabilisations across increasing radii.

However, it is important to note that all th efindings were derived from the feature importance analysis performed only for the Radius-based neighbour selection. Therefore, a similar analysis for the Node-Based neighbour selection remains an interesting avenue for further study. Given the structural differences in neighbour selection methods, it could be useful to investigate how feature importance behaves in a purely network-constrained spatial approach and whether it exhibits similar stabilisations or fluctuations at varying scales.

Moreover, it is important to clarify that low feature importance values do not necessarily mean that the corresponding features are unimportant in real-world UHI processes. Rather, it indicates that under the given spatial configuration and data characteristics, the model was not able to capture useful variation from those features. This can be due to spatial uniformity introduced through averaging at certain radii, or due to the specific distribution of a feature within this dataset. In other cities or with different data, these same features might show stronger relevance. This reinforces the interpretation that observed importance reflects not inherent feature value, but rather its predictive capacity under a specific modelling context and spatial scale.

5.2 Computational Efficiency and Model Optimisation

Random Forest (RF) proved to be the most computationally efficient and reliable modelling approach, consistently delivering high predictive performance across different experimental setups. The model converged promptly, requiring only 6-7 generations of hyperparameter tuning to reach optimal configurations. Unlike neural networks, which are highly sensitive to hyperparameter selection and require extensive computational resources for fine-tuning, RF remained relatively stable across different parameter configurations. This makes it a practical choice for large-scale urban heat island (UHI) modelling, where accuracy and computational cost must be balanced. Moreover, this also potentially means that the RF models could be used without a time-consuming GA optimisation step and still deliver competitive predictions.

An analysis of hyperparameter trends across different feature aggregation methods and neighbour selection strategies reveals several key insights:

1. Across all models, the best-performing configurations favoured tree depths between 30 and 50, suggesting that while some depth is necessary to capture nonlinear interactions, excessive depth does not provide additional benefits.

2. The number of estimators (trees) was consistently at the upper limit of the predefined range, which was set at 150-200. This suggests that increasing the number of trees further might have led to even better performance, though the initial assumption of the GA parameters range and computational constraints prevented testing beyond this range.

3. The choice of 'max_features' fluctuated between 'sqrt' and 'log2', indicating that different spatial scales required different subsets of features to optimise performance.

4. For Weighted Means, the parameters α and β define the balance between network-based and geographic-based weighting, ensuring that $\alpha + \beta = 1$. When $\alpha = 1$, the weighting is fully based on graph proximity, while when $\alpha = 0$, it is fully determined by the geographic distance between segments.

   On the one hand, for the radius-based neighbour selection, a clear patern is observed, where at the smaller radii values, α is in the neighbourhood of 1, while it diminishes to 0.25 (with some fluctuations) after $R = 300m$. This suggests that while the network effects play a meaningful role in most predictions, it is especially relevant on shorter distances. A potential reason for this might be the fact that the network-based approach resolves the challenge of excluding the streets that are geograpically close to a given segment but have, for example, a natural barrier preventing heat exchange.

   On the other hand, the node-based neighbour selection shows a less distinct pattern in fluctuations between α and $\beta = 1$. Nevertheless, it emerges from the resuls that for most predictions, geographic proximity

played a more pronounced role. Therefore, although some pattern can be traced, this relationships should be examined further.

One of the main findings is the diminishing returns of increasing the spatial radius or node count for neighbour selection using the Weighted Means feature aggregation. For the radius-based neighbour selection, at 1000m, the best-performing model reached an R^2 of 0.95, which was already close to the highest accuracy observed. Expanding the radius to 6000m, which significantly increased computational time, resulted in only a marginal improvement to 0.9586. Given the non-linear growth in computational cost at larger radii, including the entire city in the prediction process is not justified by the small gains in accuracy.

Additionally, an R^2 of approximately 0.93 was achieved at just 500m, reinforcing the idea that most relevant information for UHI modelling is contained within a relatively small spatial window. This raises the question of whether expanding to much larger scales is necessary, as the computational burden outweighs the predictive benefits.

Given this trade-off, the choice of feature aggregation method also plays a role. At smaller spatial scales, Separate Means and Weighted Means performed similarly. Since Separate Means is more straightforward to implement and requires less computation, it may be preferable for applications where efficiency is a priority. However, Weighted Means is better suited when capturing broader spatial dependencies is necessary.

These findings emphasise the importance of optimising the spatial radius for neighbour selection and choosing an aggregation method that balances accuracy and efficiency. In practical applications, for the Apeldoorn data set used in the current research, a spatial radius of 2000m-3000m or a node count of 16-18 provides the best trade-off between computational feasibility and predictive performance.

5.3 Testing the Pipeline

The pipeline developed for this study was tested on datasets from Rotterdam and Montreal in addition to the bike-collected version of Apeldoorn data. The two key takeaway are that the pipeline is well-tranferable across different data sets and that the optimal neighborhood radius is city-dependent. The best results for Rotterdam were achieved at approximately 9000m, whereas Montreal exhibited peak performance at around 1000m. This difference is likely due to several urban characteristics that differ Montreal and Rotterdam. For example, beyound the city size, Montreal has a denser urban form than Rotterdam. Moreover, the city layout and the trend of changes in urban morphology are likey different in two cities. It can be expected that the cities where the land-use and urban morphology change gradually could tend to perform better at higher radii, whereas the cities with faster change in urban form may perform better at lower radii. This could be caused by the fact that if the change of urban form happens at a larger scale, the model also needs a larger radii at the neighbour selection step to learn these patterns.

Compared to the Apeldoorn dataset used for the development of the pipeline, the datasets for Rotterdam and Montreal lacked certain key features, such as Predominant Facade—which was identified as highly important in Apeldoorn—as well as bike and pedestrian ratios. When applying the pipeline to these cities, the highest prediction performance was lower than in Apeldoorn. This suggests that these missing features contained valuable information that contributed significantly to predictive accuracy. Additionally, an attempt was made to apply a model trained on Apeldoorn to test predictions in Montreal and Rotterdam, but it did not yield meaningful results. This confirms that the model needs to be trained on each city's dataset separately rather than assuming direct generalisability across different urban environments.

A separate experiment was conducted using bike dataset measurements. The bike dataset only included a limited selection of street segments forming a closed loop. This experimental setup tested whether sampling a single neighbour at varying distances would meaningfully improve prediction accuracy. On the one hand, the experiment with a SUHI type of UHI suggested that the dataset with to have only several neighbours at different spatial scale is not sufficient for a meaningful prediction enhancement. On the other hand, a similar experiment with CUHI showed the opposite results, which should be further investigated.

An important observation from the canopy temperature predictions was that it had a significantly higher performance than the land surface temperature simulation. In fact, the performance of the canopy model was close to the results obtained for Apeldoorn satellite data with significantly more street segments in the training set. This might be connected to the fact that the relative humidity of the segment emerged as the most critical predictor. This significantly higher predictive accuracy may stem from an inherent correlation between canopy temperature and relative humidity, potentially leading to a form of recursive prediction.

Ideally, environmental characteristics that are directly correlated with the target variable should not be included in training, as they introduce dependencies that can inflate prediction accuracy without truly capturing the underlying processes. For example, if mean surface temperature were included in the prediction of the temperature

differential (ΔT), the model's accuracy would approach 100% since ΔT can be derived directly from surface temperature and reference temperature. While this hypothesis needs to be further examined, it is important to carefully select the input variables to avoid circular dependencies.

Additionally, part of the day was not included in the training process due to its categorical nature. However, prior research by Pena Acosta [3] identified this feature as having high predictive importance. Its exclusion may have negatively impacted model performance, and future work should explore strategies to effectively incorporate time-of-day variations into predictive models. From this retrospective, the pipeline could have been tested with three different datasets, morning, afternoon, and evening. On top of the potential enhancement of the prediction, it yields insights into variations in the results during the transitions between different times of the day.

5.4 Recommendations

Several areas for further refinement and investigation have been identified:

1. The dataset split for training was set at 20%, whereas many studies suggest using a 30% to 35% split. Testing different train-test ratios could provide insights into whether a smaller training set leads to decreased performance, which would however further prevent potential overfitting.

2. A more systematic and robust approach to neural networks should be developed, as the current study found them to be suboptimal compared to Random Forests. Further tuning and architectural modifications may enhance their predictive ability.

3. The sensitivity of the model to weight type remains unclear. While the intermittent pattern observed between alpha and beta suggests variability, it is not known how much performance differs between cases where, for example, $\alpha = 1$, $\beta = 0$ versus $\alpha = 0.75$, $\beta = 0.25$. A more focused sensitivity analysis could quantify the impact of different weighting schemes. The observed fluctuations in the optimal alpha values may indicate that the differences in performance between weighting configurations are relatively minor, making the choice of α and β somewhat stochastic.

4. More cities should be tested using the node-based neighbour selection and different α and β proportions. In Montreal, Rotterdam, and the bike data for Apeldoorn, this was not feasible due to the fragmentation of the street segments, which disrupted network connectivity. This limitation, however, also highlights a key drawback of the node-based neighbour selection compared to the radius-based approach.

5. The proposed pipeline could be simplified. If the difference between alpha and beta contributions is found to be marginal, then using only distance-based weights would be more computationally efficient and easier to implement. Additionally, since fragmented data (as in Rotterdam and Montreal) can only be processed using radius-based neighbour selection, relying solely on a distance-based neighbour selection would still achieve high predictive performance while eliminating the need to construct a graph. This simplified version of the model would rely solely on distance neighbour selection and feature aggregation, which involves a simple sequence of calculating distances between all street segments, followed by selecting neighbours based on radius thresholds, and weighting features based on a precomputed distance matrix. Note that the sequence of operations above does not require constructing a graph structure, which can significantly simplify and speed up the process. Given these benefits, this streamlined version of the model is recommended as a baseline for future research. As Leonardo da Vinci once said, "Simplicity is the ultimate sophistication."

6 Conclusion

This research has explored the impact of explicitly incorporating spatial dependencies in Urban Heat Island (UHI) modelling, demonstrating its substantial contribution to improving predictive accuracy. By addressing spatial relationships directly within machine learning frameworks, it was shown that models are better equipped to capture localised variations in urban temperature distributions.

The study has discovered that an optimal spatial scale exists for UHI prediction, which varies between cities. This highlights the necessity of city-specific calibration when implementing spatial models. Regarding spatial neighbour selection, the analysis compared node-based and radius-based methods. While radius-based neighbour selection consistently exhibited superior performance and broader applicability, further investigation is required.

In terms of feature aggregation, Separate Means and Weighted Means methods were examined. The results indicate that separate means are computationally more efficient and particularly suitable for smaller spatial scales, whereas weighted means provide the highest predictive accuracy and robustness across different scales.

A diverse set of machine learning models, including linear regression, ensemble methods, and deep learning architectures, was evaluated. Among these, Random Forest (RF) emerged as the best-performing model, offering a strong balance between interpretability and predictive power.

By systematically integrating spatial dependencies, tuning spatial scale, refining neighbour selection, and optimising feature aggregation, this thesis provides a refined framework for enhancing UHI modelling. Future work should focus on refining the radius-based neighbour selection, improving computational efficiency for large-scale applications, and further exploring deep learning approaches to push predictive performance even further.

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A Classification of ML models

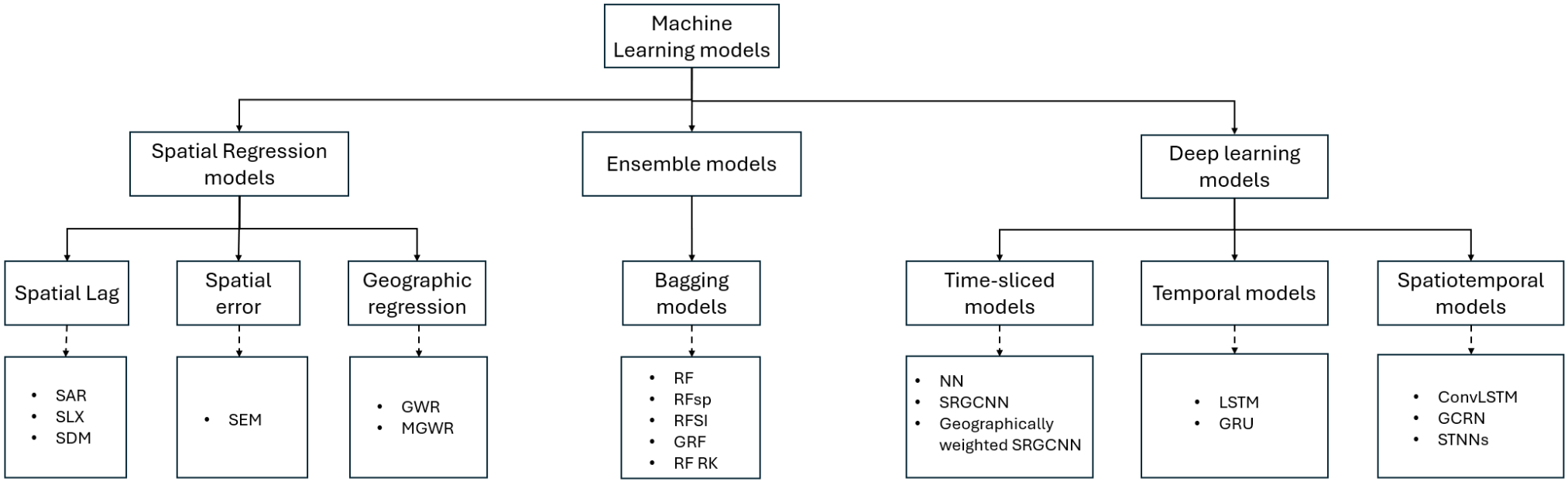

Figure 24: Classification of ML models

B Genetic Algorithm Results

For each feature aggregation method, the hyperparameters corresponding to the GA optimisation results were found per radius or number of nodes. The current appendix presents these results for various datasets.

B.1 Apeldoorn Satellite data

The results for the Separate Means neighbour selection method can be seen in Table 12.

Table 12: Best Hyperparameters per Radius for Separate Means neighbour selection

Radius [m]	Best R^2	n_estimators	max_features	max_depth	min_samples_split	bootstrap
100	0.7762	200	sqrt	100	2	False
200	0.8483	200	log2	50	2	False
300	0.8881	50	sqrt	100	4	False
400	0.9100	150	log2	50	4	False
500	0.9224	200	sqrt	50	2	False
600	0.9278	50	log2	30	2	False
700	0.9297	200	log2	50	2	False
1000	0.9373	100	sqrt	50	4	False
2000	0.9474	100	sqrt	30	2	False
3000	0.9480	100	log2	30	2	False
4000	0.9414	200	log2	30	2	False
5000	0.9264	150	log2	50	2	False
6000	0.8731	100	sqrt	50	2	False
7000	0.7245	50	log2	100	4	False
8000	0.6853	200	log2	30	2	False
8500	0.6794	200	log2	30	2	False
8700	0.6782	200	sqrt	30	2	False

Similarly, for the United Means, the best hyperparameters per radius can be seen in Table 13.

Table 13: Best Hyperparameters per Radius for the United Means

Radius [m]	Best R^2	n_estimators	max_features	max_depth	min_samples_split	bootstrap
100	0.7537	200	log2	30	2	False
200	0.7971	200	sqrt	30	2	False
300	0.8333	200	sqrt	30	2	False
400	0.8544	150	sqrt	100	2	False
500	0.8558	100	sqrt	30	2	False
600	0.8526	150	log2	50	2	False
700	0.8650	200	sqrt	50	2	False
1000	0.8708	200	log2	30	2	False
2000	0.8847	100	log2	30	4	False
3000	0.8989	200	log2	30	2	False
4000	0.8860	150	sqrt	100	2	False
5000	0.8811	200	sqrt	30	2	False
6000	0.8414	50	log2	30	4	False
7000	0.7199	200	sqrt	30	2	False
8000	0.6813	200	sqrt	100	2	False
8500	0.6734	200	log2	100	4	False
8700	0.6744	150	log2	100	2	False

For the PCA Means, see Table 14.

Table 14: Best Hyperparameters per Radius for PCA Means

Radius [m]	Best R^2	n_estimators	max_features	max_depth	min_samples_split	bootstrap
100	0.7415	200	log2	50	2	False
200	0.7898	150	sqrt	100	2	False
300	0.8213	150	log2	100	4	False
400	0.8445	200	sqrt	30	2	False
500	0.8454	200	sqrt	30	2	False
600	0.8506	200	sqrt	30	2	False
700	0.8622	200	log2	30	2	False
1000	0.8682	200	log2	30	2	False
2000	0.8838	150	sqrt	30	2	False
3000	0.9001	150	sqrt	30	2	False
4000	0.8881	100	sqrt	30	2	False
5000	0.8816	150	log2	50	2	False
6000	0.8430	200	log2	50	2	False
7000	0.7232	150	sqrt	100	4	False
8000	0.6801	200	sqrt	50	4	False
8500	0.6755	200	sqrt	50	2	False
8700	0.6753	200	log2	50	2	False

Finally, the result for the Weighed Mean neighbours selection with an additional α hyperparameter can be seen in Table 15.

Table 15: Best Hyperparameters per Radius for Weighted Means

Radius [m]	Best R^2	Alpha	n_estimators	max_features	max_depth	min_samples_split	bootstrap
100	0.7568	0.75	100	log2	100	2	False
200	0.8460	1.00	200	sqrt	30	2	False
300	0.8869	0.50	150	log2	30	2	False
400	0.9095	0.25	150	sqrt	100	2	False
500	0.9271	0.25	100	sqrt	30	2	False
600	0.9352	0.25	200	log2	100	2	False
700	0.9405	0.25	200	log2	100	2	False
1000	0.9500	0.25	200	log2	50	2	False
2000	0.9585	0.50	200	sqrt	30	2	False
3000	0.9582	0.00	200	log2	100	2	False
4000	0.9585	0.00	150	log2	50	2	False
5000	0.9578	0.00	200	log2	30	2	False
6000	0.9586	0.25	150	log2	50	2	False
7000	0.9569	0.25	100	sqrt	100	2	False
8000	0.9560	0.25	100	sqrt	30	2	False
8500	0.9562	0.25	100	log2	100	2	False
8700	0.9552	0.25	200	sqrt	100	2	False

Analogously, the feature aggregation method and the hyperparameters corresponding to the GA optimisation results were found per number of nodes as a search threshold. The results for the Separate Means neighbour selection method can be seen in Table 16.

Table 16: Best Hyperparameters per Node Count for Separate Means (Node-Based Neighbour Selection)

Nodes	Best R^2	n_estimators	max_features	max_depth	min_samples_split	bootstrap
0	0.7307	200	sqrt	100	2	False
1	0.7826	200	sqrt	30	2	False
2	0.8221	150	sqrt	30	2	False
4	0.8692	200	log2	30	2	False
6	0.8905	150	log2	30	2	False
8	0.9003	100	log2	30	2	False
10	0.9082	200	log2	100	2	False
12	0.9136	200	log2	50	2	False
14	0.9135	150	sqrt	30	2	False
16	0.9163	100	sqrt	50	2	False
18	0.9152	200	sqrt	50	2	False
20	0.9151	150	sqrt	30	2	False
36	0.9108	100	sqrt	50	4	False
52	0.9050	150	sqrt	30	2	False
68	0.8912	200	log2	100	4	False
84	0.8842	200	sqrt	30	4	False
100	0.8772	150	log2	30	4	False
125	0.8567	200	log2	50	2	False
150	0.7370	150	sqrt	100	2	False
175	0.7038	50	log2	100	2	False
200	0.6832	200	sqrt	100	2	False
205	0.6831	200	log2	30	2	False
210	0.6779	200	log2	50	2	False
215	0.6793	200	sqrt	100	2	False

Similarly, for the United Means, the best hyperparameters per radius can be seen in Table 17.

Table 17: Best Hyperparameters per Node Count for United Means (Node-Based Neighbour Selection)

Nodes	Best R^2	n_estimators	max_features	max_depth	min_samples_split	bootstrap
0	0.7211	200	log2	30	2	False
1	0.7626	200	log2	100	2	False
2	0.7829	200	log2	100	2	False
4	0.8146	150	log2	30	2	False
6	0.8344	200	log2	50	2	False
8	0.8457	200	log2	50	2	False
10	0.8473	200	sqrt	100	4	False
12	0.8569	200	sqrt	50	2	False
14	0.8626	100	sqrt	100	2	False
16	0.8597	200	log2	100	2	False
18	0.8595	200	log2	100	4	False
20	0.8566	100	log2	30	2	False
36	0.8592	200	sqrt	100	2	False
52	0.8645	200	sqrt	30	4	False
68	0.8477	100	sqrt	50	2	False
84	0.8368	150	sqrt	50	2	False
100	0.8371	50	sqrt	30	6	False
125	0.8199	150	sqrt	30	2	False
150	0.7388	150	sqrt	30	2	False
175	0.7002	200	sqrt	100	4	False
200	0.6801	200	sqrt	30	2	False
205	0.6796	200	log2	30	2	False
210	0.6770	150	sqrt	100	2	False
215	0.6718	200	sqrt	50	4	False

For the PCA Means, see Table 18.

Table 18: Best Hyperparameters per Node Count for PCA Means (Node-Based Neighbour Selection)

Nodes	Best R^2	n_estimators	max_features	max_depth	min_samples_split	bootstrap
0	0.7171	200	sqrt	100	2	False
1	0.7428	200	sqrt	50	2	False
2	0.7684	200	log2	30	2	False
4	0.8007	200	sqrt	30	2	False
6	0.8154	100	log2	30	2	False
8	0.8308	200	log2	30	2	False
10	0.8425	200	log2	30	2	False
12	0.8533	200	log2	30	2	False
14	0.8577	200	sqrt	50	2	False
16	0.8553	200	log2	50	4	False
18	0.8599	150	sqrt	30	2	False
20	0.8557	200	sqrt	30	2	False
36	0.8579	100	sqrt	100	4	False
52	0.8659	200	log2	30	2	False
68	0.8489	200	log2	30	4	False
84	0.8321	200	sqrt	50	2	False
100	0.8298	200	sqrt	100	2	False
125	0.8102	200	sqrt	30	2	False
150	0.7321	100	sqrt	100	2	False
175	0.6950	150	sqrt	50	2	False
200	0.6786	200	sqrt	100	2	False
205	0.6771	150	log2	30	4	False
210	0.6774	150	log2	30	2	False
215	0.6726	200	sqrt	50	2	False

Finally, the result for the Weighed Mean neighbours selection with an additional α hyperparameter can be seen in Table 19.

Table 19: Best Hyperparameters per Node Count for Weighted Means (Node-Based Neighbour Selection)

Nodes	Best R^2	Alpha	n_estimators	max_features	max_depth	min_samples_split	bootstrap
0	0.7280	1.00	200	sqrt	100	4	False
1	0.7689	0.00	200	sqrt	100	2	False
2	0.7991	0.25	150	log2	50	4	False
4	0.8649	0.25	200	sqrt	100	2	False
6	0.8972	0.25	200	log2	50	2	False
8	0.9093	0.25	200	log2	30	2	False
10	0.9166	0.25	200	log2	30	4	False
12	0.9271	0.25	150	log2	50	2	False
14	0.9297	0.75	200	log2	50	2	False
16	0.9307	0.25	200	log2	100	2	False
18	0.9287	0.25	100	log2	20	2	False
20	0.9327	0.25	200	sqrt	100	2	False
36	0.9360	0.25	150	sqrt	50	4	False
52	0.9404	0.50	200	sqrt	50	4	False
68	0.9453	0.25	100	sqrt	100	2	False
84	0.9441	0.25	150	log2	30	2	False
100	0.9430	0.25	150	sqrt	20	2	False
125	0.9485	0.25	200	log2	30	2	False
150	0.9533	0.00	50	sqrt	30	2	False
175	0.9540	0.00	200	log2	50	2	False
200	0.9535	0.00	50	sqrt	50	2	False
205	0.9546	0.25	200	sqrt	30	2	False
210	0.9543	0.25	200	sqrt	50	2	False
215	0.9551	0.00	100	sqrt	30	2	False

B.2 Rotterdam Satellite data

The results for the Separate Means neighbour selection method can be seen in Table 20 while for the Weighted Means, the best hyperparameters per radius can be seen in Table 21.

Table 20: Best Hyperparameters per Radius for Experiment (Separate Means)

Radius [m]	Best R^2	n_estimators	max_features	max_depth	min_samples_split	bootstrap
100	0.4611	100	sqrt	30	6	False
200	0.6302	200	sqrt	50	2	False
300	0.7151	150	log2	30	6	False
400	0.7486	150	sqrt	100	2	False
500	0.7677	200	log2	20	2	False
750	0.8276	200	log2	30	2	False
1000	0.8501	200	log2	30	2	False
1250	0.8609	200	log2	50	2	False
1500	0.8682	150	sqrt	100	2	False
1750	0.8690	200	log2	50	2	False
2000	0.8781	200	sqrt	50	4	False
2500	0.8748	150	sqrt	20	4	False
3000	0.8726	50	log2	30	6	False
3500	0.8737	200	log2	100	2	False
4000	0.8551	200	log2	30	4	False
5000	0.8719	100	sqrt	100	4	False
7000	0.8812	150	log2	30	2	False
9000	0.8484	150	log2	50	2	False
11000	0.8357	100	sqrt	100	4	False
13000	0.7961	200	sqrt	20	4	False
15000	0.7792	150	log2	20	6	False
20000	0.7681	50	log2	50	4	False
30000	0.7264	100	log2	30	6	False
43000	0.3411	150	log2	100	4	True

Table 21: Best Hyperparameters per Radius for Experiment (Weighted Means)

Radius [m]	Best R^2	n_estimators	max_features	max_depth	min_samples_split	bootstrap
0	0.3460	200	sqrt	10	4	False
100	0.4819	100	sqrt	100	2	True
200	0.5891	200	sqrt	30	2	False
300	0.6963	100	log2	100	2	False
400	0.7483	200	log2	100	2	False
500	0.7710	150	sqrt	30	4	False
750	0.8320	100	log2	50	2	False
1000	0.8596	200	sqrt	20	2	False
1250	0.8768	150	sqrt	20	4	False
1500	0.8863	200	log2	20	2	False
1750	0.8931	50	sqrt	50	2	False
2000	0.8945	200	sqrt	20	4	False
2500	0.8989	200	log2	100	2	False
3000	0.8985	100	sqrt	30	2	False
3500	0.9026	200	log2	100	2	False
4000	0.8948	200	sqrt	20	2	False
5000	0.8994	150	log2	30	2	False
7000	0.9013	150	log2	50	2	False
9000	0.9098	200	sqrt	50	2	False
11000	0.9081	150	sqrt	30	2	False
13000	0.9075	200	sqrt	30	2	False
15000	0.9071	100	log2	100	2	False
20000	0.8984	200	sqrt	100	4	False
30000	0.8842	150	sqrt	20	2	False
43000	0.8910	150	log2	20	2	False

B.3 Montreal Satellite data

The hyperparameters corresponding to the GA optimisation results per radius for the Separate Means neighbour selection method can be seen in Table 22.

Table 22: Best Hyperparameters per Radius for Separate Means - Montreal

Radius [m]	Best R^2	n_estimators	max_features	max_depth	min_samples_split	bootstrap
100	0.6647	150	sqrt	20	2	False
200	0.7550	200	log2	50	6	False
300	0.7932	100	sqrt	50	4	False
400	0.8145	150	sqrt	30	2	False
500	0.8353	100	sqrt	100	2	False
750	0.8375	200	log2	20	2	False
1000	0.8307	200	log2	20	4	False
1500	0.8331	100	sqrt	30	2	False
2000	0.8313	200	log2	100	2	False
3000	0.8188	200	sqrt	50	4	False
4000	0.8137	200	sqrt	30	4	False
5000	0.8219	150	sqrt	100	4	False
7000	0.8319	200	log2	50	4	False
10000	0.6865	50	sqrt	30	2	False
13000	0.5737	200	sqrt	50	6	False
14500	0.5405	150	sqrt	50	2	False

For the Weighted Means, the best hyperparameters per radius can be seen in Table 23.

Table 23: Best Hyperparameters per Radius for Weighted Means - Montreal

Radius [m]	Best R^2	n_estimators	max_features	max_depth	min_samples_split	bootstrap
0	0.5250	150	sqrt	30	2	True
100	0.6306	150	log2	50	2	False
200	0.7355	200	sqrt	100	2	False
300	0.7851	150	sqrt	50	2	False
400	0.8180	200	log2	30	2	False
500	0.8338	150	sqrt	50	4	False
750	0.8579	200	log2	30	2	False
1000	0.8614	150	sqrt	100	4	False
1500	0.8613	100	log2	100	2	False
2000	0.8534	50	log2	50	2	False
3000	0.8652	200	sqrt	50	2	False
4000	0.8536	150	log2	50	2	False
5000	0.8447	200	sqrt	100	4	False
7000	0.8441	200	log2	100	2	False
10000	0.8516	150	sqrt	100	2	False
13000	0.8408	200	log2	50	4	False
14500	0.8417	150	sqrt	20	2	False

B.4 Apeldoorn Bike data

The detailed results for the SUHI type of UHI including best hyperparameters for both used feature aggregation methods are summarised in Tables 24 and 25

Table 24: Best Hyperparameters per Radius for the Surface Temperature Experiment (Separate Means)

Radius [m]	Best R^2	n_estimators	max_features	max_depth	min_samples_split	bootstrap
100	0.7833	150	log2	30	2	False
250	0.7711	100	log2	100	4	False
500	0.7873	50	log2	100	2	False
750	0.7920	150	sqrt	30	2	False
1000	0.7865	150	log2	20	4	False
1500	0.7950	200	sqrt	30	4	False
2000	0.7939	200	log2	50	2	False
3000	0.7969	200	sqrt	50	4	False
4000	0.7986	200	log2	30	2	False
4500	0.7838	100	sqrt	100	2	False
5140	0.7555	30	sqrt	30	2	False

Table 25: Best Hyperparameters per Radius for the Surface Temperature Experiment (Weighted Means)

Radius [m]	Best R^2	n_estimators	max_features	max_depth	min_samples_split	bootstrap
0	0.7736	200	sqrt	30	2	False
100	0.7814	150	log2	50	2	False
250	0.7661	150	log2	20	2	False
500	0.7772	150	log2	30	2	False
750	0.7873	150	log2	30	2	False
1000	0.7917	150	sqrt	100	2	False
1500	0.7939	150	sqrt	50	2	False
2000	0.7936	100	log2	100	2	False
3000	0.7986	150	sqrt	30	2	False
4000	0.7975	150	log2	50	2	False
4500	0.7970	100	log2	100	2	False
5140	0.7947	150	log2	100	2	False

Similarly, for the simulations where Canopy Temperature was set as as a target, the detailed results including best hyperparameters for both Separate Means and Weighted Means feature aggregation method are summarised in Tables 24 and 25

Table 26: Best Hyperparameters per Radius for the Canopy Experiment (Separate Means)

Radius [m]	Best R^2	n_estimators	max_features	max_depth	min_samples_split	bootstrap
100	0.8876	200	sqrt	50	2	False
250	0.9024	200	sqrt	30	2	False
500	0.9196	100	sqrt	50	6	False
750	0.9230	150	sqrt	50	4	False
1000	0.9252	200	log2	100	4	False
1500	0.9249	150	log2	100	2	False
2000	0.9205	100	log2	50	4	False
3000	0.9247	150	log2	100	2	False
4000	0.9184	150	sqrt	30	4	False
4500	0.9213	200	sqrt	30	4	False
5140	0.9204	150	log2	100	2	False

Table 27: Best Hyperparameters per Radius for the Canopy Experiment (Weighted Means)

Radius [m]	**Best R^2**	**n_estimators**	**max_features**	**max_depth**	**min_samples_split**	**bootstrap**
0	0.8345	200	sqrt	30	2	False
100	0.8863	200	sqrt	100	2	False
250	0.9015	150	sqrt	50	4	False
500	0.9236	200	log2	30	2	False
750	0.9263	200	log2	100	2	False
1000	0.9282	200	sqrt	30	2	False
1500	0.9271	100	log2	100	2	False
2000	0.9276	200	sqrt	50	2	False
3000	0.9249	100	log2	50	2	False
4000	0.9239	50	log2	50	2	False
4500	0.9246	150	log2	100	4	False
5140	0.9276	150	log2	30	2	False